

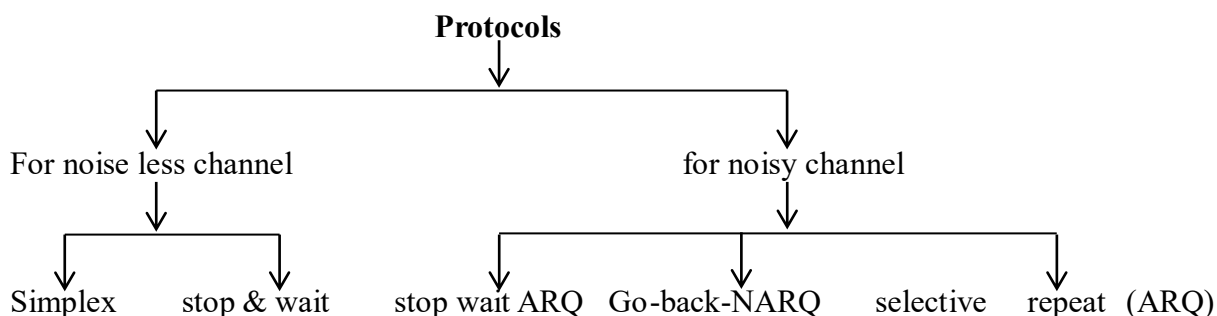
UNIT-II

(Elementary data link protocols: simplex protocol, A simplex stop and wait protocol for an error-free channel, A simplex stop and wait protocol for noisy channel. Sliding Window protocols: A one-bit sliding window protocol, A protocol using Go-Back-N, A protocol using Selective Repeat, Example data link protocols. Medium Access sub layer: The channel allocation problem, Multiple access protocols: ALOHA, Carrier sense multiple access protocols, collision free protocols. Wireless LANs, Data link layer switching.)

Elementary Data Link Layer Protocols:

A protocol is a set of rules that govern data communications and it represents an agreement between the communicating devices.

The protocols that can be used for noiseless (error – free) channels and those that can be used for noisy (error – creating) channels.



At the protocols that are Uni-directional in the sense that the data frames travel from one node called sender to another node called receiver.

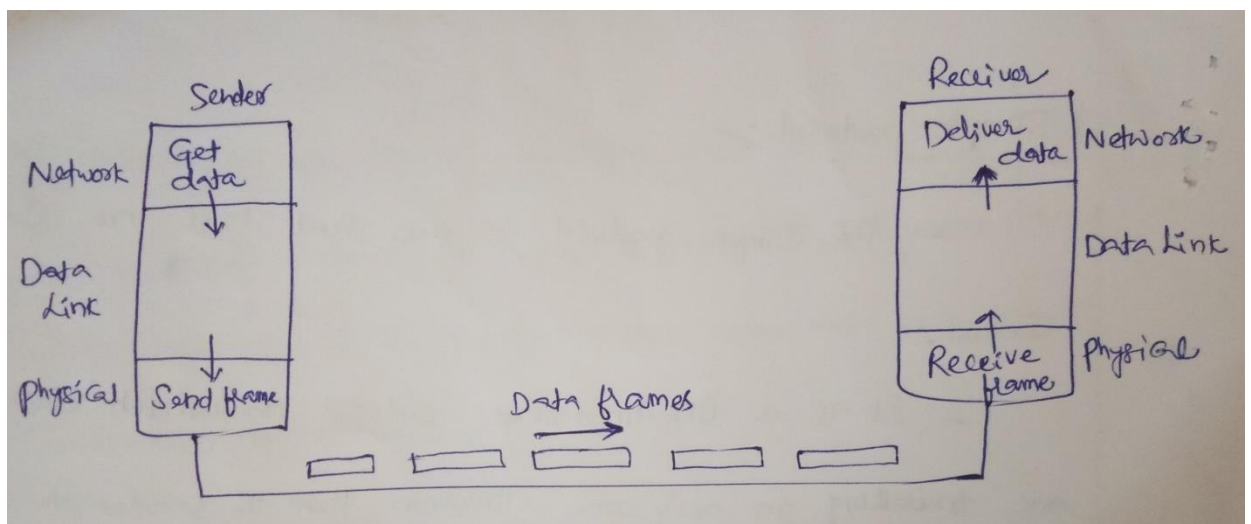
The special frames are called acknowledgement (ACK). Negative acknowledgement (NAK) can flow in the opposite direction for flow and error control purposes, data flow in only one direction.

Simplex Protocol:

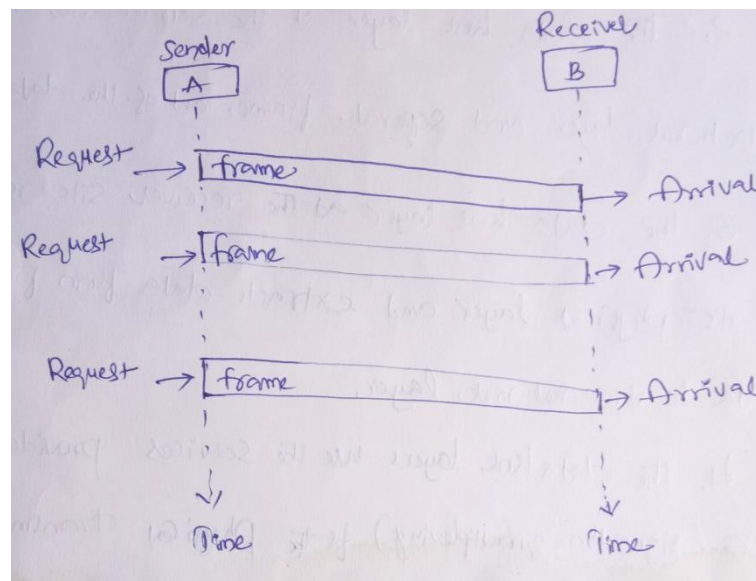
The simplex protocol is one that has no flow (or) error control. It is a unidirectional protocol in which data frames are travelling in only one directional from the sender to receiver. Assume that the receiver can immediately handle any frame it receives with a processing time that is small enough and negligible. The data link layer of the receiver immediately removes the header from the frames and takes the data packet to its network layer.

The design issues are given by for simple protocol is

1. There is no need for flow control process.
2. The data link layer at the sender site gets data from its network layer and separate frame out of the data and send it
3. The data link layer at the receiver site receives a frame from its physical layer and extracts data from frame and delivers the data to its network layer.
4. The data link layers use the services provided by their physical layers (signaling and multiplexing) for the physical transmission of bits.



5. The sender site cannot send a frame until its network layer has a data packets to send and the receiver site cannot deliver a data packet to its network layer until a frame arrives.
6. Suppose, if the protocol is implemented as a procedure, introduce an event in the protocol. The procedure at the sender site is also constantly running there is no action until a request from the network layer.
7. The procedure at the receiver site is also constantly running, but there is no action until notification from the physical layer arrives.

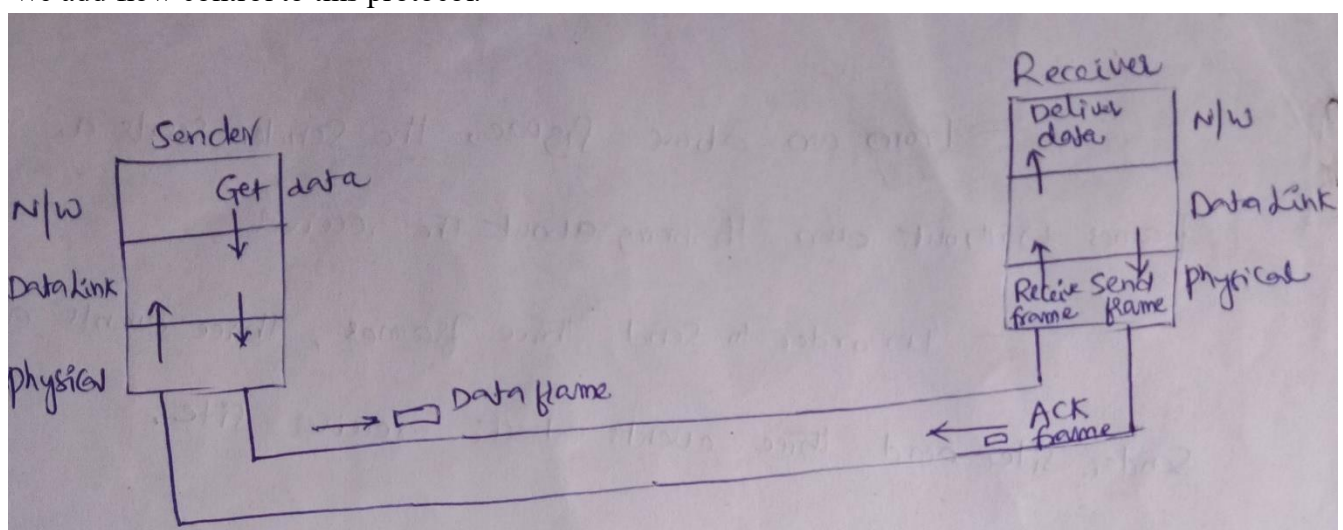


From above figure, the sender sends a sequence of frames without even thinking about the receiver. In order to send three frames, three events occur at the sender site and three events at the receiver site.

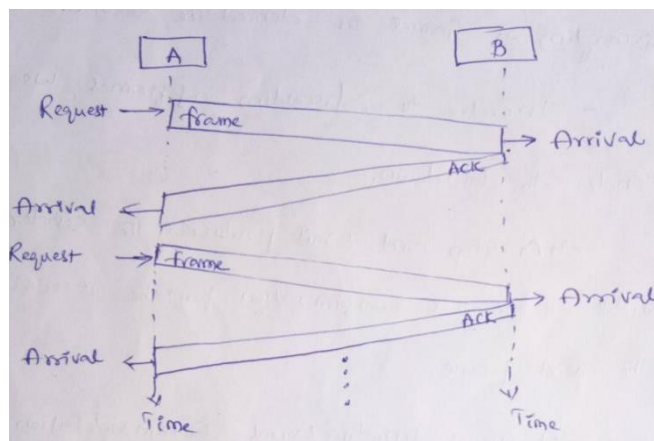
Stop – And – Wait Protocol:

If the data frames arrive at the receiver site faster than they can be processed, the frames must be stored until their use. The receiver does not have enough storage space, especially it is receiving data from many sources, which may result in either the discarding of frames or denied the service. In order to discarding of frames we need to tell the sender to slowdown. In a stop – and – wait protocol, the sender sends one frame stops until it receives confirmation from the receiver and then sends the next frame. It is a unidirectional communication for data frames, but auxiliary ACK frames (acknowledgment) travel from other direction.

We add flow control to this protocol.



From an above figure, at any time there is either one data frame on the forward channel or one ACK frame on the reverse channel. The link is like a half-duplex



The sender sends one frame and waits for feedback from the receiver. When acknowledgement arrives, the sender sends the next frame. Note: sending two frames in the protocol involves the sender in four events and receiver in two events.

Noisy Channel Protocols

Sliding Window Protocol:

Sliding window is one of the method of error correction. To increase the data rate, this method allows the sender to transmit a specific number of packets in continuous mode. i.e., at the maximum possible rate. The no. of packets that can be transmitted in such a way that i.e. window size. Window size can be constant parameter, which means that it is chosen at connection setup and does not change entire session. If the destination receives the packet with corrupted data, it might send a negative acknowledgement (NAKK), specifying that the packet needs to be retransmitted. When a data frame arrives, instead of immediately sending a separate control frame, the receiver restrains itself and waits until the network layer passes it the next packet.

Sender Window Size

It is a list of consecutive frame sequence numbers that can be sent by the sender or that have been sent and acknowledgment are waited for. When an ACK arrives and all previous frames have already been acknowledged the window can be transmitted with next highest available sequence number

Receiver Window Size

It is a list of sequence numbers for frames that can be accepts by the receiver. When a valid frame arrives and previous frames have already arrived the window is advanced. If a frame arrive that is not with in the 'window' is discarded.

Advantages

1. Sliding window is simpler, having only one set of parameters to manage.
2. Simultaneous communication in both direction is possible
3. Better utilization if n/w band width especially if there are large transmission delays.
4. Traffic flow with reverse traffic data, known as piggy backing.

Stop & Wait ARQ:

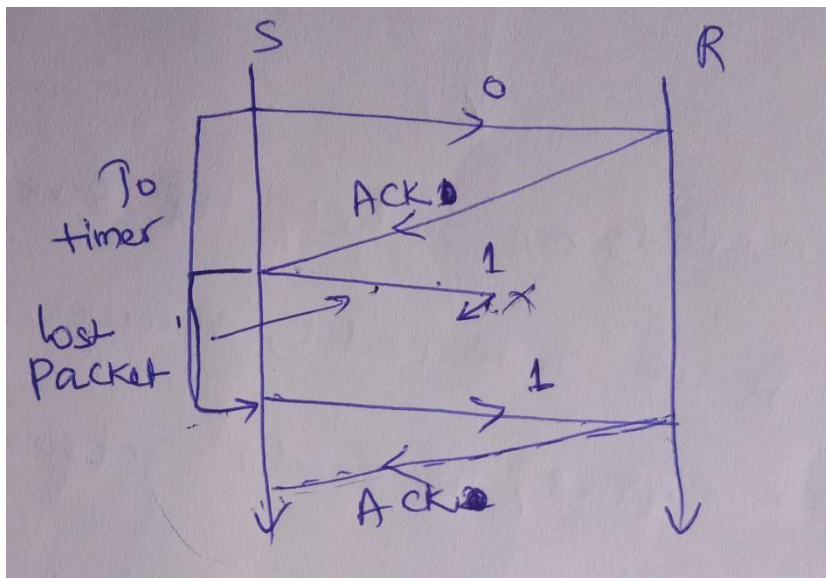
This is the simplex protocol with sequence numbers and with ACK frame indicating the sequence number of the next frame expected. In this sliding protocol, the maximum window size is 1. Such a protocol uses stop & wait protocol since the sender transmits a frame and waits for its acknowledgement before sending to the next one.

One – bit sliding window protocol is also called stop wait – ARQ.

The following issues, which may occur in stop and wait protocol are

1. Data packet lost
2. Acknowledgement lost
3. Delayed acknowledgement

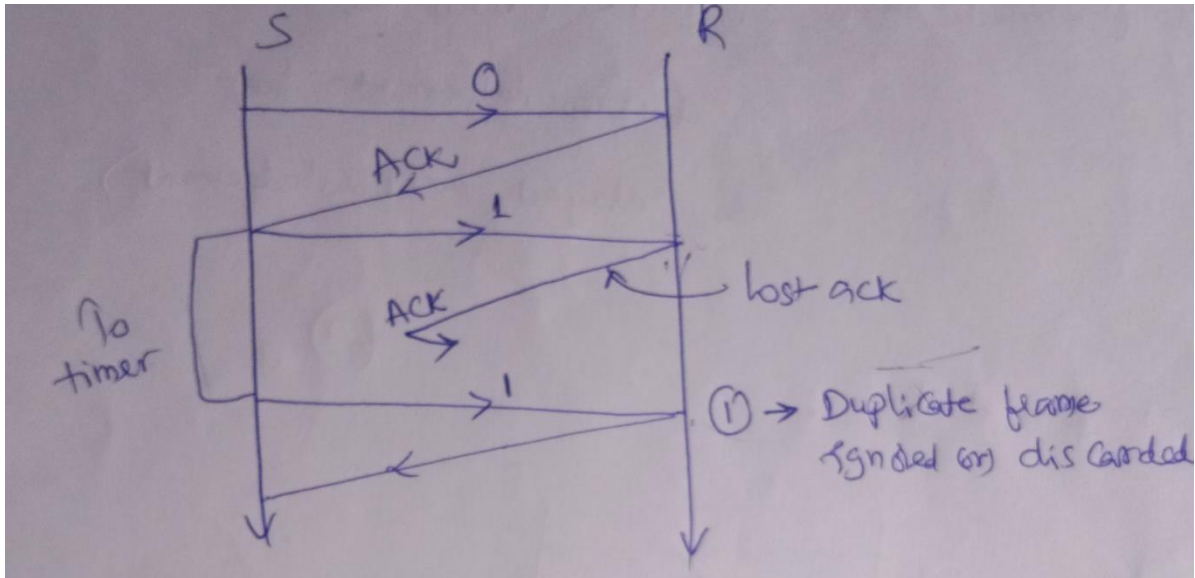
Data Packet Lost



In a above data packet lost, when a receiver receives the frame and found it damaged or lost, it is discarded but retains its number, when sender does not receive its acknowledgement it retransmits the same frame, when timer expires. ("To → Time out times for retransmitting of packets).

Usually a timer is set by sender after each frame is transmitted, its acknowledgement must be received before timer expires.

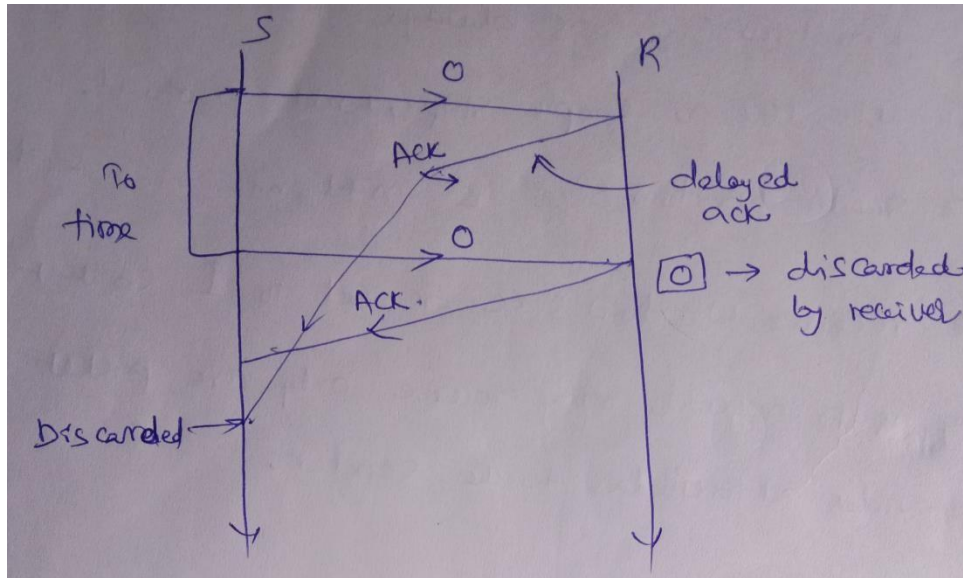
Acknowledgement Lost



When an acknowledgement is lost the sender does not know whether the frame is received by receiver. As for the timer expires, the sender retransmitted the same frame. On the other hand, receiver has already received this frame earlier hence sender copy of the frame is discarded. So the sender will think about sending the data packets in frames by providing sequence numbers.

Delayed Acknowledgement

The ACK frame may be delayed due to some link problem. The ACK is received after the time r is elapsed. While the sender has already transmitted the same frame Again second ACK is initiated by receiver for the transmitted frame, hence the second ACK is discarded.



In order to avoid duplication the acknowledgment numbered. (S & W + To + sequence number to acknowledgements).

Go Back N (GBN) Protocol:

Go back N uses the sliding window flow control protocol. If no errors occurs the operations are identical to sliding window. A station may send multiple frames as allowed by the window size. The Go Back N follows the methods to approach error force transmission with frames /packet.

1. Sender window size ($W_S > 1$)
2. Receiver window size ($W_R = 1$)
3. Acknowledgements (cumulative & independent acknowledgements)

Sender Window Size

The sender window is always greater than one, otherwise it acts like a simple stop & wait protocol.

The sender window size is $n+1, n+2, \dots$ etc

Receiver window size: is set to 1 so that the capability of the receiver may receive only one packet at a time provides acknowledgment to the sender.

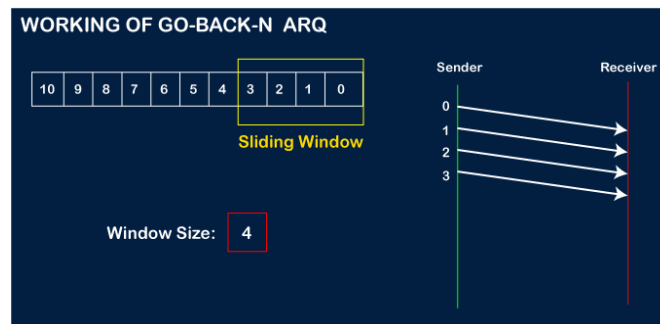
- The sequence number of the outbound frames depends upon the size of the sender's window. N is the sender's window size.
- If the size of the sender's window is 4 then the sequence number will be 0,1,2,3,0,1,2,3,0,1,2, and so on.

The number of bits in the sequence number is 2 to generate the binary sequence 00,01,10,11.

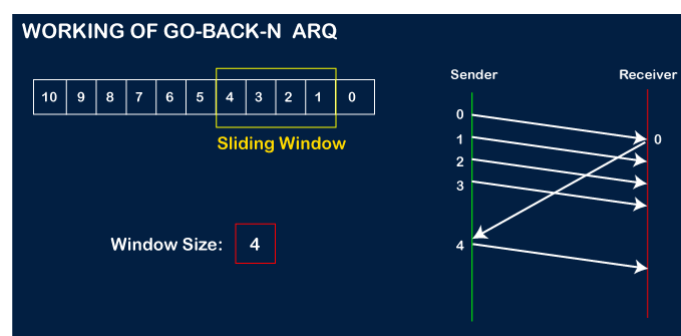
Working of Go-Back-N ARQ

Suppose there are a sender and a receiver, and let's assume that there are 11 frames to be sent. These frames are represented as 0,1,2,3,4,5,6,7,8,9,10, and these are the sequence numbers of the frames. Let's consider the window size as 4, which means that the four frames can be sent at a time before expecting the acknowledgment of the first frame.

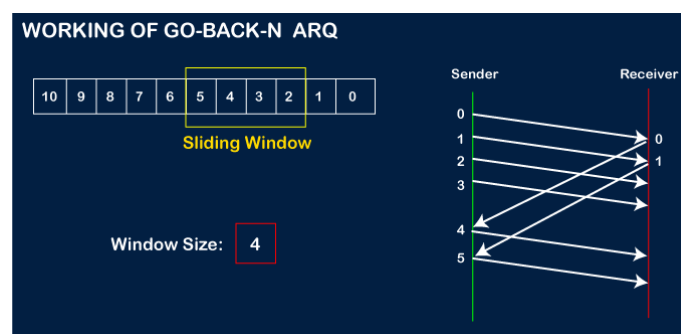
Firstly, the sender will send the first four frames to the receiver, i.e., 0,1,2,3, and now the sender is expected to receive the acknowledgment of the 0th frame.



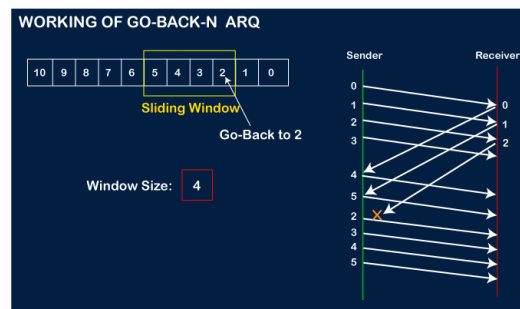
Let's assume that the receiver has sent the acknowledgment for the 0 frame, and the receiver has successfully received it. The sender will then send the next frame, i.e., 4, and the window slides containing four frames (1,2,3,4).



The receiver will then send the acknowledgment for the frame no 1. After receiving the acknowledgment, the sender will send the next frame, i.e., frame no 5, and the window will slide having four frames (2,3,4,5).



Now, let's assume that the receiver is not acknowledging the frame no 2, either the frame is lost, or the acknowledgment is lost. Instead of sending the frame no 6, the sender Go-Back to 2, which is the first frame of the current window, retransmits all the frames in the current window, i.e., 2,3,4,5.



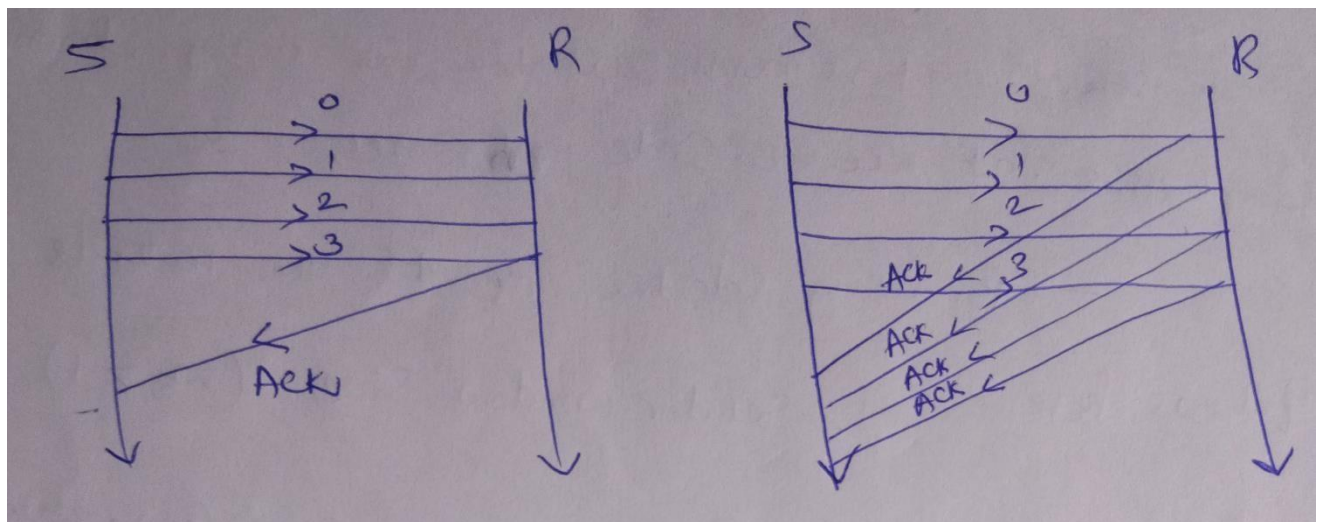
Acknowledgements

Acknowledgements are provided in the go back N are

1. Cumulative acknowledgement
2. Independent acknowledgement

Go Back N always use cumulative acknowledgement

Cumulative Acknowledgments & Independent Acknowledgment:



In cumulative acknowledgements, the sender may send more packets in a slot the receiver only and one particular acknowledgement to the sender. The advantage of this cumulative acknowledgement is less traffic. The drawback of this less reliable, because any one packet must be lost whole transmission from sender is collapsed.

In independent acknowledgement, for every packet there is an acknowledgement. The advantage of this independent acknowledgement is more reliable, because any of the

packets may be lost, the sender sends remaining packets without any discrepancy. The drawback of this model is high traffic.

Selective Repeat (SR) Protocol:

The selective repeat ARQ is the model in which retransmits only the damaged packets instead of sending multiple frames.

The selective retransmission increases the efficiency of transmission and is more suitable for a noisy channel. The circuit complexities at the receiver side increase.

By using selective repeat the methods which may follow are:

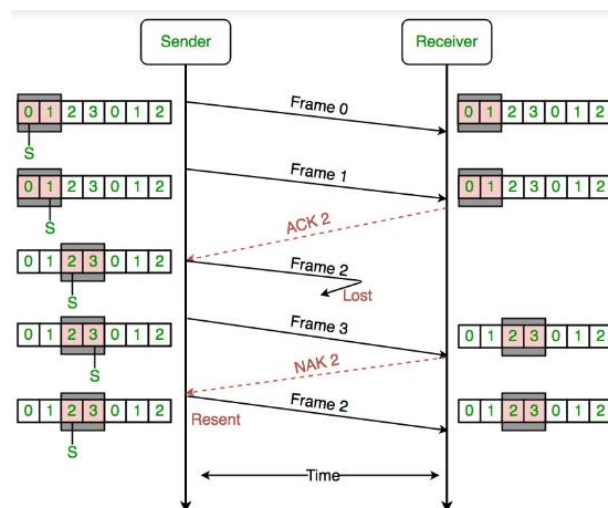
1. Sender window size ($W_S > 1$)
2. Receiver window size ($W_R = W_S$)
3. Acknowledgements (Negative acknowledgements)

Sender Window Size:

The sender window size is always greater than that of the selective repeat. In order to transmit more packets on the sender side.

Receiver Window Size:

The receiver window is equal to the sender window size as the receiver simply follows the sender to acknowledge the incoming packets.

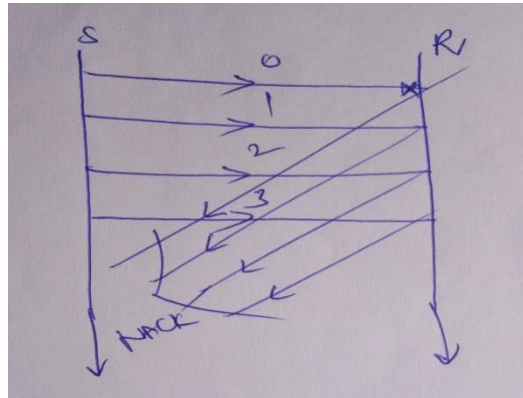


From an above figure, if the packet (0) is lost while sending receiver could not accept the packet and it accepts the packets (1,2,3) in the receiver window reference.

The lost packet is retransmitted when timer expires, during this event we transmit only the lost packet selectively. So this is called selective repeat (SR).

Acknowledgements:

So that the receiver which may provide negative acknowledgement for lost packet for retransmission.



In this negative acknowledgement the sender will assume that packet may lost and retransmit again. In any stop & wait protocol, the transmit time is more than the acknowledgment time for successful packets delivery in order to improve the efficiency.

Difference between Go-Back-N and Selective Repeat Protocol

S.NO	Go Back 'N'	Selective Repeat
1	In Go-Back-N Protocol, if the sent frame are found suspected then all the frames are re-transmitted from the lost packet to the last packet transmitted.	In selective Repeat protocol, only those frames are re-transmitted which are found suspected.
2	Sender window size of Go-Back-N Protocol is N.	Sender window size of selective Repeat protocol is also N.
3	Receiver window size of Go-Back-N Protocol is 1.	Receiver window size of selective Repeat protocol is N.
4	Go-Back-N Protocol is less complex.	Selective Repeat protocol is more complex.
5	In Go-Back-N Protocol, neither sender nor at receiver need sorting.	In selective Repeat protocol, receiver side needs sorting to sort the frames.
6	In Go-Back-N Protocol, type of Acknowledgement is cumulative.	In selective Repeat protocol, type of Acknowledgement is individual.
7	In Go-Back-N Protocol, Out-of-Order packets are NOT Accepted (discarded) and the entire window is re-transmitted.	In selective Repeat protocol, Out-of-Order packets are Accepted.
8	In Go-Back-N Protocol, if Receiver receives a corrupt packet, then also, the entire window is re-transmitted.	In selective Repeat protocol, if Receiver receives a corrupt packet, it immediately sends a negative acknowledgement and hence only the selective packet is retransmitted.
9	Efficiency of Go-Back-N Protocol is $N/(1+2*a)$	Efficiency of selective Repeat protocol is also $N/(1+2*a)$

Both Go-Back-N Protocol and Selective Repeat Protocol are the types of sliding window protocols.

The main difference between these two protocols is, that after finding the suspect or damage in sent frames go-back-n protocol re-transmits all the frames, whereas selective repeat protocol re-transmits only that frame which is damaged.

Example Data Link protocols

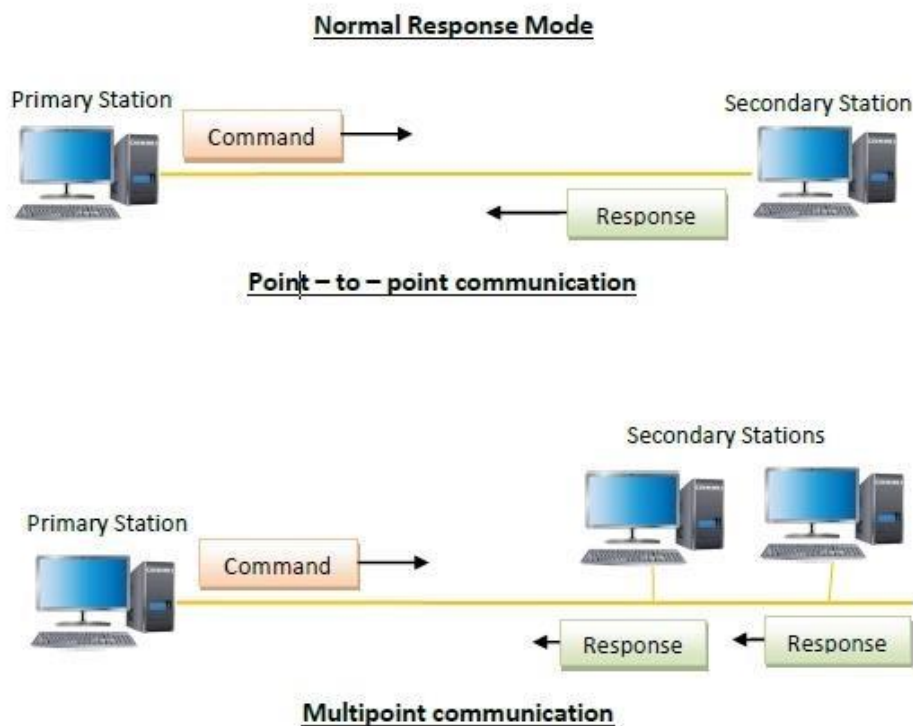
HDLC (High-level Data Link Control)

High-level Data Link Control (HDLC) is a group of communication protocols of the data link layer for transmitting data between network points or nodes. Since it is a data link protocol, data is organized into frames. A frame is transmitted via the network to the destination that verifies its successful arrival. It is a bit - oriented protocol that is applicable for both point - to - point and multipoint communications.

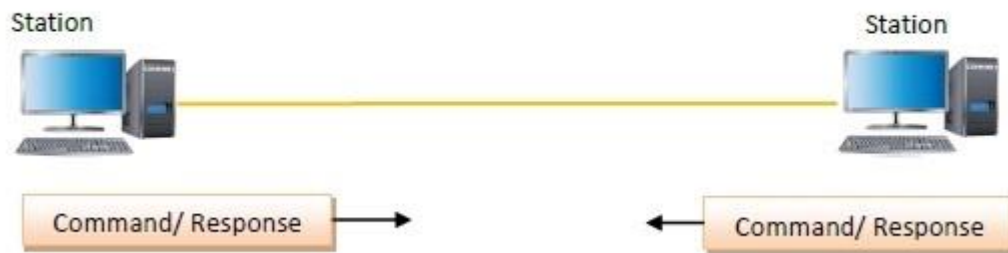
Transfer Modes

HDLC supports two types of transfer modes, normal response mode and asynchronous balanced mode.

- **Normal Response Mode (NRM)** – Here, two types of stations are there, a primary station that send commands and secondary station that can respond to received commands. It is used for both point - to - point and multipoint communications.

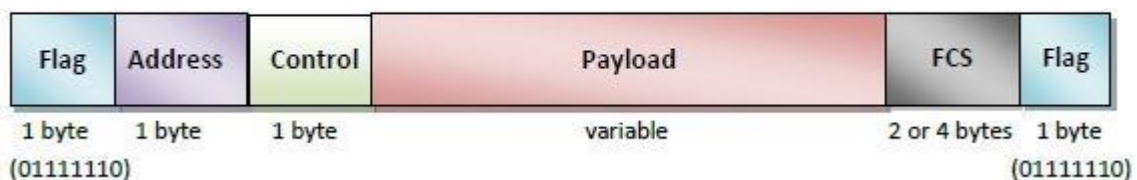


- **Asynchronous Balanced Mode (ABM)** – Here, the configuration is balanced, i.e. each station can both send commands and respond to commands. It is used for only point - to - point communications.

Asynchronous Balanced Mode**HDLC Frame**

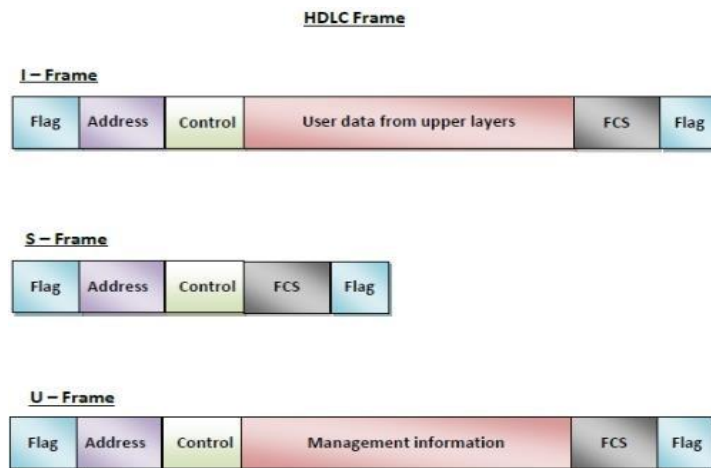
HDLC is a bit - oriented protocol where each frame contains up to six fields. The structure varies according to the type of frame. The fields of a HDLC frame are –

- **Flag** – It is an 8-bit sequence that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.
- **Address** – It contains the address of the receiver. If the frame is sent by the primary station, it contains the address(es) of the secondary station(s). If it is sent by the secondary station, it contains the address of the primary station. The address field may be from 1 byte to several bytes.
- **Control** – It is 1 or 2 bytes containing flow and error control information.
- **Payload** – This carries the data from the network layer. Its length may vary from one network to another.
- **FCS** – It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)

HDLC Frame**Types of HDLC Frames**

There are three types of HDLC frames. The type of frame is determined by the control field of the frame –

- **I-frame** – I-frames or Information frames carry user data from the network layer. They also include flow and error control information that is piggybacked on user data. The first bit of control field of I-frame is 0.
- **S-frame** – S-frames or Supervisory frames do not contain information field. They are used for flow and error control when piggybacking is not required. The first two bits of control field of S-frame is 10.
- **U-frame** – U-frames or Un-numbered frames are used for myriad miscellaneous functions, like link management. It may contain an information field, if required. The first two bits of control field of U-frame is 11.



PPP (Point - to - Point Protocol)

Point - to - Point Protocol (PPP) is a communication protocol of the data link layer that is used to transmit multiprotocol data between two directly connected (point-to-point) computers. It is a byte - oriented protocol that is widely used in broadband communications having heavy loads and high speeds. Since it is a data link layer protocol, data is transmitted in frames. It is also known as RFC 1661.

Services Provided by PPP

The main services provided by Point - to - Point Protocol are –

- Defining the frame format of the data to be transmitted.
- Defining the procedure of establishing link between two points and exchange of data.
- Stating the method of encapsulation of network layer data in the frame.
- Stating authentication rules of the communicating devices.
- Providing address for network communication.
- Providing connections over multiple links.
- Supporting a variety of network layer protocols by providing a range OS services.

Components of PPP

Point - to - Point Protocol is a layered protocol having three components –

- **Encapsulation Component** – It encapsulates the datagram so that it can be transmitted over the specified physical layer.
- **Link Control Protocol (LCP)** – It is responsible for establishing, configuring, testing, maintaining and terminating links for transmission. It also imparts negotiation for set up of options and use of features by the two endpoints of the links.
- **Authentication Protocols (AP)** – These protocols authenticate endpoints for use of services. The two authentication protocols of PPP are –

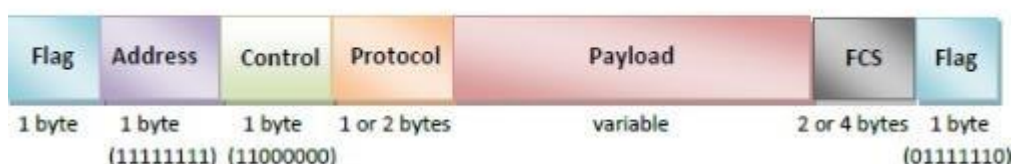
- Password Authentication Protocol (PAP)
- Challenge Handshake Authentication Protocol (CHAP)
- **Network Control Protocols (NCPs)** – These protocols are used for negotiating the parameters and facilities for the network layer. For every higher-layer protocol supported by PPP, one NCP is there. Some of the NCPs of PPP are –
 - Internet Protocol Control Protocol (IPCP)
 - OSI Network Layer Control Protocol (OSINLCP)
 - Internetwork Packet Exchange Control Protocol (IPXCP)
 - DECnet Phase IV Control Protocol (DNCP)
 - NetBIOS Frames Control Protocol (NBFCP)
 - IPv6 Control Protocol (IPV6CP)

PPP Frame

PPP is a byte - oriented protocol where each field of the frame is composed of one or more bytes. The fields of a PPP frame are –

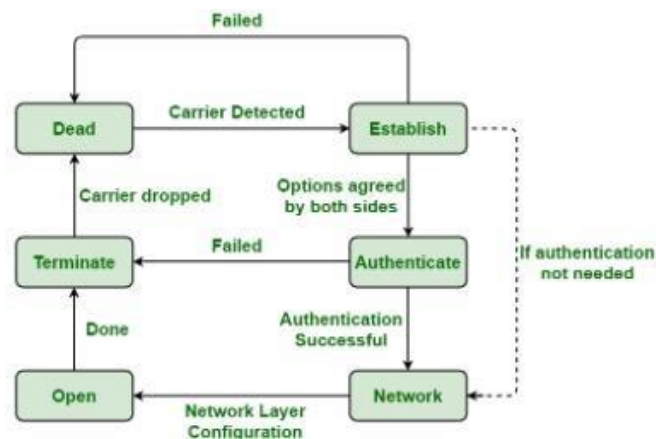
- **Flag** – 1 byte that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.
- **Address** – 1 byte which is set to 11111111 in case of broadcast.
- **Control** – 1 byte set to a constant value of 11000000.
- **Protocol** – 1 or 2 bytes that define the type of data contained in the payload field. □ **Payload** – This carries the data from the network layer. The maximum length of the payload field is 1500 bytes. However, this may be negotiated between the endpoints of communication.
- **FCS** – It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)

PPP Frame



Byte Stuffing in PPP Frame – Byte stuffing is used in PPP payload field whenever the flag sequence appears in the message, so that the receiver does not consider it as the end of the frame. The escape byte, 01111101, is stuffed before every byte that contains the same byte as the flag byte or the escape byte. The receiver on receiving the message removes the escape byte before passing it onto the network layer.

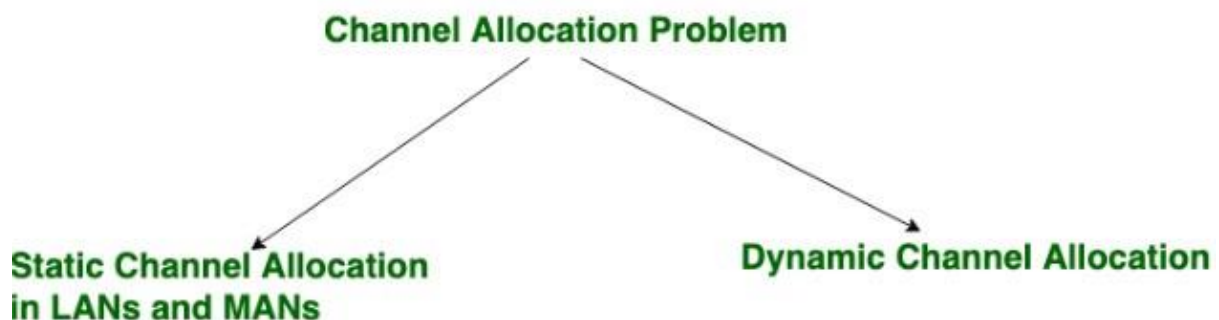
PPP Transition Phases



Channel Allocation Problem

Channel allocation is a process in which a single channel is divided and allotted to multiple users in order to carry user specific tasks. There are user's quantity may vary every time the process takes place. If there are N number of users and channel is divided into N equalized sub channels, Each user is assigned one portion. If the number of users are small and don't vary at times, then Frequency Division Multiplexing can be used as it is a simple and efficient channel bandwidth allocating technique.

Channel allocation problem can be solved by two schemes: Static Channel Allocation in LANs and MANs, and Dynamic Channel Allocation.



Static Channel Allocation

The traditional way of allocating a single channel, among multiple competing users is to chop up its capacity by using one of the multiplexing such as FDM. If there are N users, the bandwidth is divided into N equal-sized portions, with each user being assigned one portion. Since each user has a private frequency band, there is now no interference among users. When there is only a small and constant number of users, each of which has a steady stream of traffic, this division is a simple and efficient allocation mechanism. A wireless example is FM radio stations.

- Static Channel Allocation Techniques Two common static channel allocation techniques are TDMA and FDMA. Time Division Multiple Access (TDMA) – With TDMA the time axis is divided into time slots of a fixed length. Each user is allocated a fixed set of time slots at which it can transmit. TDMA requires that users be synchronized to a common clock. Typically extra overhead bits are required for synchronization.
- Frequency Division Multiple Access (FDMA) – With FDMA the available frequency bandwidth is divided into disjoint frequency bands. A fixed band is allocated to each user. FDMA requires a guard band between user frequency bands to avoid cross-talk. Another static allocation technique is Code Division Multiple Access (CDMA), this technique is used in many wireless networks.

The performance of static channel allocation depends on:

1. The variation in the number of users over time
2. The nature of the traffic sent by the user

If the traffic on a shared medium is from a fixed number of sources each transmitting at a fixed rate, static channel allocation can be very efficient.

Problems with Static Channel Allocation

- However, when the number of senders is large and varying or the traffic is bursty, FDM presents some problems. If the spectrum is cut up into N regions and fewer than N users are currently interested in communicating, a large piece of valuable spectrum will be wasted.
- And if more than N users want to communicate, some of them will be denied permission for lack of bandwidth, even if some of the users who have been assigned a frequency band hardly ever transmit or receive anything.
- A static allocation is a poor fit to most computer systems, in which data traffic is extremely bursty.

To avoid problems associated with static allocation, one to think about dynamic allocation which perform better.

Dynamic Channel Allocation:

Dynamic channel allocation is schemes for allotting shared network channels to competing users in a dynamic manner as per their requirements. There are a number of methods for dynamic channel allocation.

The key assumptions of these methods are –

Independent Traffic – It is assumed that the users are independent of each other, i.e. each user has to send or receive data which is independent of the data transmission of the other users in the network system. Each user has a program for generating frames for transmission. Once the

user generates a frame for transmission, it is blocked until the frame has been successfully transmitted.

Single Channel – The algorithms assume that all contending stations request for transmission via a single channel. All stations are considered to be equally capable of transmission. However, the algorithm may assign different priorities to them for contention.

Detectable Collisions – If two frames from different stations are simultaneously transmitted, the resulting signal is distorted, and a collision is said to occur. If a collision occurs, all stations should be able to detect the collision. The frames lost due to collisions are retransmitted.

Continuous Time or Slotted Time – Depending upon the allocation scheme, time may be considered as continuous or slotted. In continuous time, frame transmission can start at any instant. In slotted time, time is divided into discrete slots. If a slot does not contain any frame, it is called an idle slot; if it contains a single frame, then the transmission is successful; if it contains more than one frames, then a collision is said to occur.

Carrier Sense or No Carrier Sense – The stations may or may not be capable of detecting whether the channel is in use before sending the frames. In algorithms which are based upon carrier sense, a station sends frame only when it senses that the channel is not busy. On the other hand, in algorithms based upon no carrier sense, the stations transmit a frame when it is available and later are informed whether successful transmission had occurred or not.

Protocol Assumption:

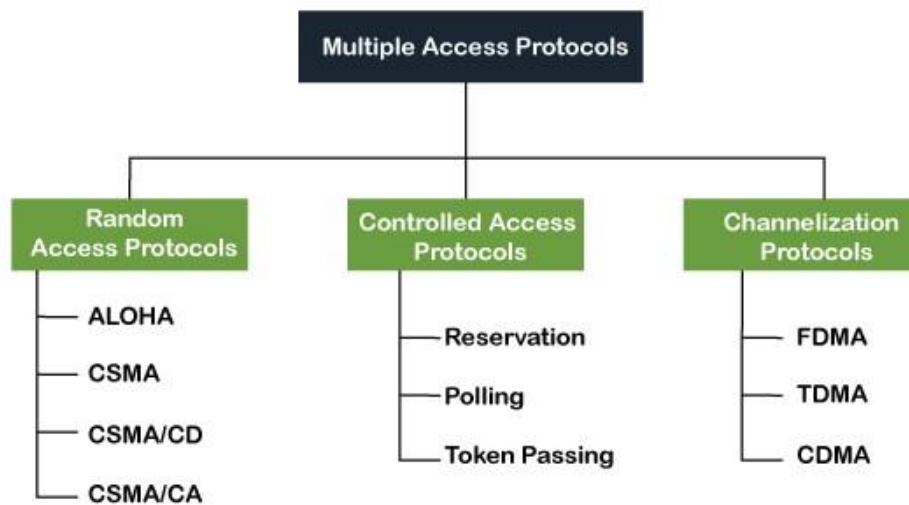
- N independent stations.
- A station is blocked until its generated frame is transmitted.
- Probability of a frame being generated in a period of length Δt is $I\Delta t$ where I is the arrival rate of frames.
- Only a single Channel available.
- Time can be either: Continuous or slotted.
- **Carrier Sense:** A station can sense if a channel is already busy before transmission. □
- **No Carrier Sense:** Time out used to sense loss data.

Multiple Access Protocol:

In LAN's the back bone is a street channel (or) transmission link, which provides all users to access the transmission facility. It may be possible that two (or) more stations transmitting simultaneously, causing their signals to interfere and becomes garbled.

In order to resolve this conflict no. of different control mechanism or access protocols have been given. Access the medium from many entry prints is called contention. It is controlled with in contention protocol. In a random access method, each station has the right to the medium without being controlled by other station. However if more

than one station tries to send, there is an access conflict. i.e. collision and the frames will be either destroyed (or) modified.



The random access techniques are given by

1. ALOHA
2. Carrier sense multiple access (CSMA)
3. CSMA/CD (CD – collision detection)

1. ALOHA:

The ALOHA protocol was developed at the university of Hawaii in the earth 1970's

ALOHA was developed for packet radio networks it is used in any shared transmission medium.

In this multiple users try to send messages to other stations through a common broadcast medium i.e., random access or contention technique.

When a station sends data, another station may attempt to do so at the same time. The data from the two stations collide and becomes garbled

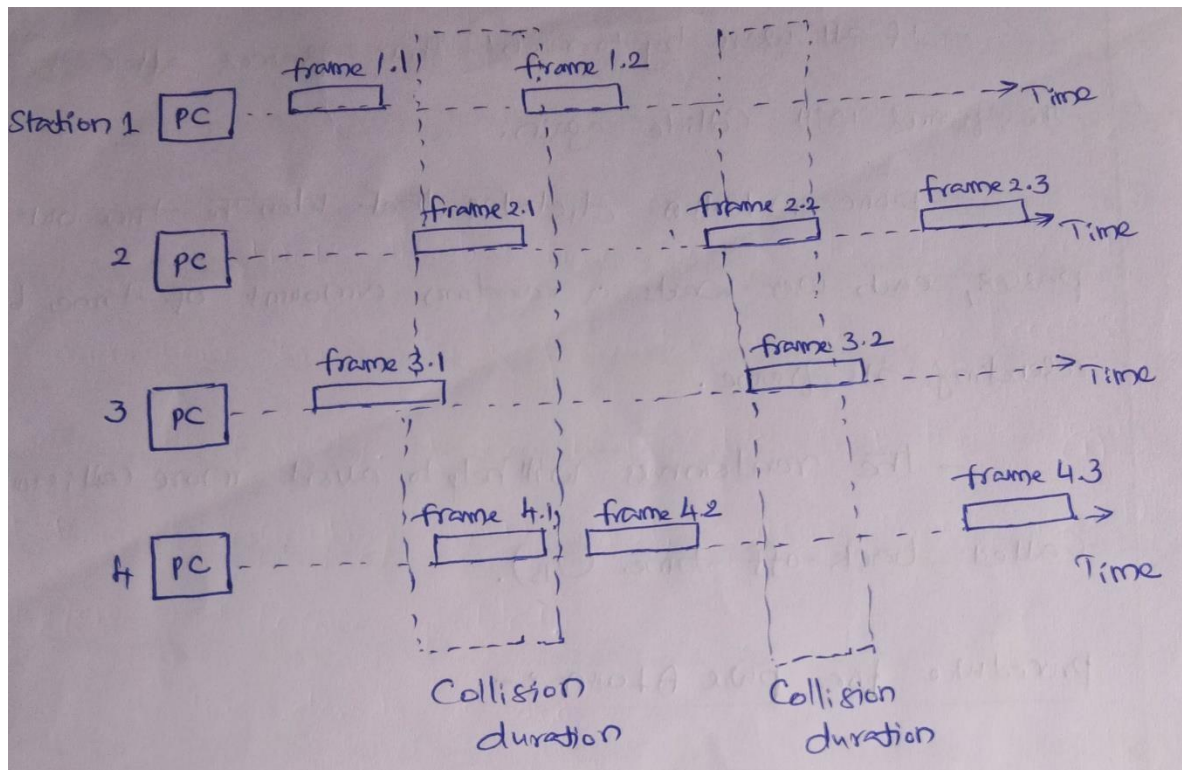
If two signals collided so, each station would simply wait a random time and try again.

PURE ALOHA:

The original ALOHA protocol is called pure ALOHA. it is a simple protocol, the idea given by the each station sends a frame wherever it has a frame to send.

Since there is only one channel to share, there is possibility of collision between frames from different stations

The below figure shows that frame collisions in pure ALOHA



The pure ALOHA relies on acknowledgements from the receiver. When a user sends a frame, it expects the receiver to send an acknowledgment. If acknowledgement does not arrive after a time out period, the station assumes that the frame has been destroyed and resends the frames. Whenever two frames try to occupy the channel at the same time, there will be collision and both will be garbled, If the first bit of a new frame overlaps with just the last bit of a frame almost finished, both frames will be totally destroyed and both will be transmitted again. If all users try to send their frames after the time out the frames will collide again.

Pure ALOHA dictates that when the time out period passes, each user waits a random amount of time before resending its frames. The randomness will help to avoid more collisions. This is called back – off time (T_B)

Procedure For Pure ALOHA:

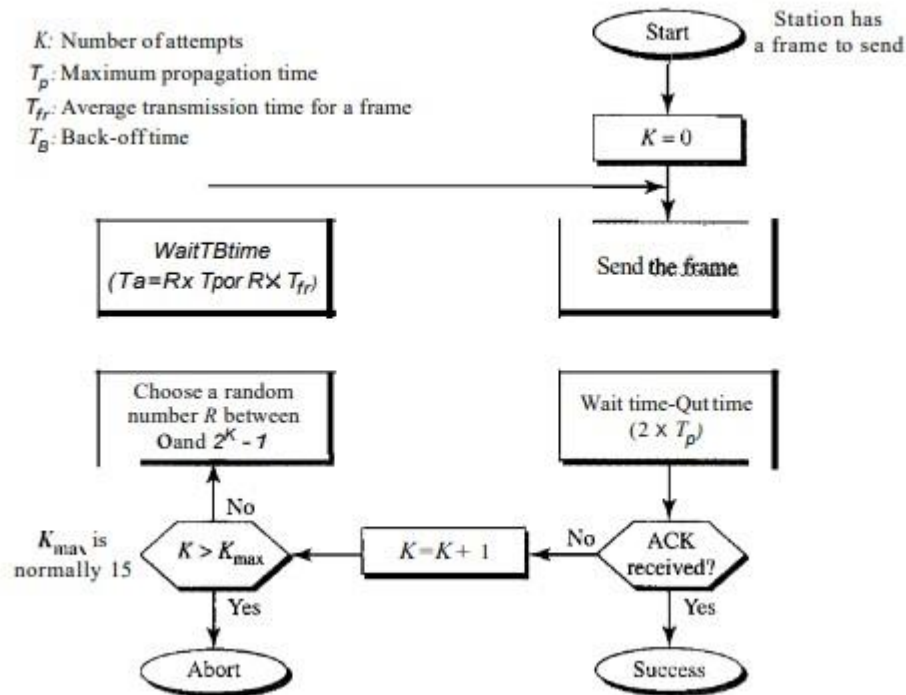
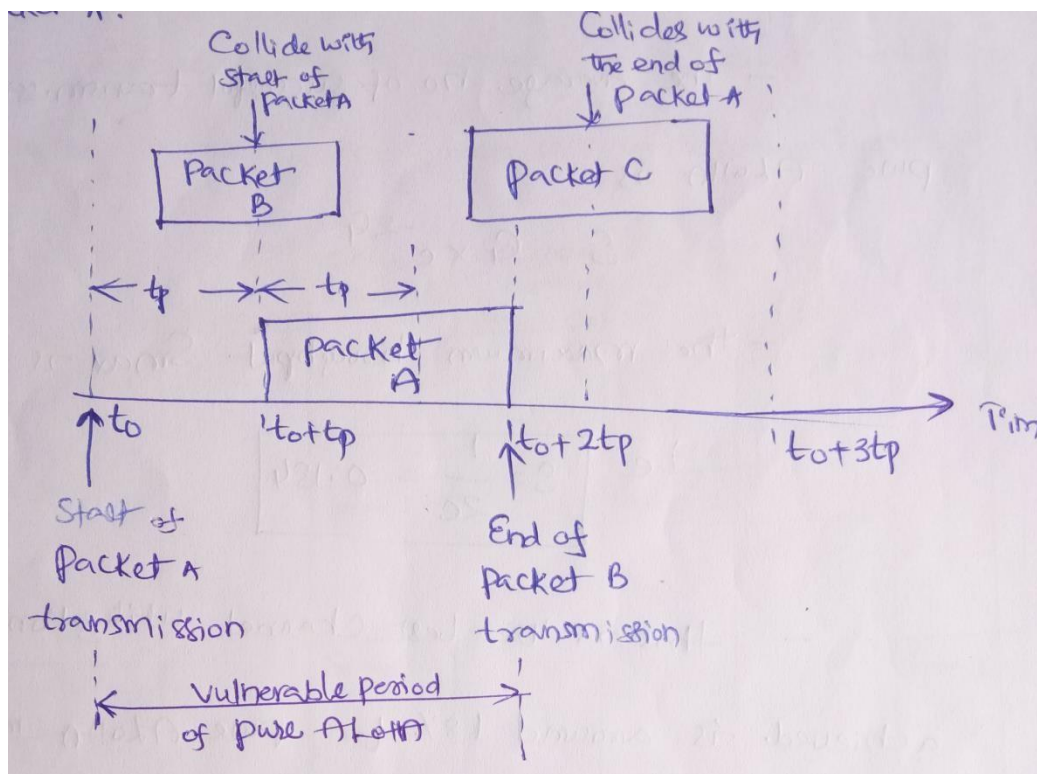


Figure: working for pure ALOHA

From an above flow chart, the time out period is equal to the maximum possible round trip propagation delay, which is twice the amount of time required to send a frame between the two most widely separated stations ($2 \times J_p$).

Let all the packets have same length and each required one time unit for transmission (t_p).

Consider any user to send a packet A at time t_0 if any other user B has generated a packet between time t_0 and $t_0 + t_p$, the end of packet B will collide with the beginning of packet A.



Since in pure ALOHA packet, a station does not listen to the channel before transmitting it has no way of knowing that above frames was already under way. Similarly if another user wants to transmit between $(t_0 + t_p)$ and $(t_0 + 2t_p)$ i.e, packet C, the beginning of packet C will collide with the end of packet A.

Thus if two packets overlap by even the smallest amount in the vulnerable period both packets will be corrupted and need to be retransmitted.

Through Put: the through put is defined as average successful traffic transmitted between stations per unit time.

The unit of time is slot – time, which is the time required to transmit a frame

The average no. Successful transmission time for pure ALOHA is

$$S = G \times e^{-2G}$$

The maximum throughput $S_{\max}$ is 0.184, for $G = \frac{1}{2}$

$$\text{i.e., } S = \frac{1}{2e} = 0.184$$

This is the best channel utilization that can be achieved is around 18% for pure ALOHA method.

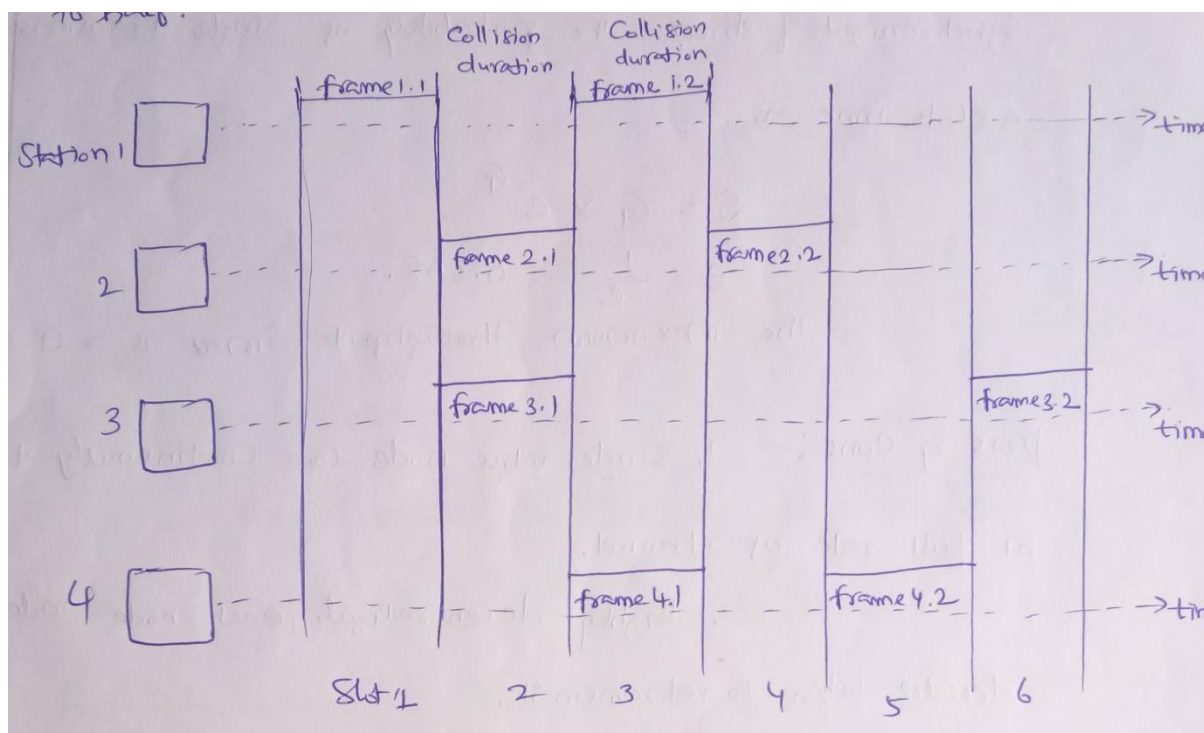
Advantages: it is a simple protocol which can result in low cost stations since no synchronization is required between stations in the each system.

Slotted ALOHA:

It was invented for improve the efficiency of pure ALOHA.

In slotted ALOHA the synchronized to these time slots and the stations are allowed to transmit at specific instance of time. All users are then synchronized to these time slots, so that whenever a user generates a packet it must synchronize exactly with the next possible channel slot.

Consequently the wasted time due to collision can be reduced to one packet time or vulnerable period is reduced to half.



Transmission attempts for four network user and random retransmission delays for colliding packets in slotted ALOHA.

In slotted ALOHA, the packets arrive in a synchronized fashion. The probability of single transmission during a slot time is

$$S = G \times e^{-G}$$

$$S = \frac{1}{e} = 0.368$$

The maximum through put $S_{\max}$ is $= 0.368$ $\therefore G = 1$

Pros & Cons:

- Single active node can continuously transmit at full rate of channel.
- Highly decentralized, each node independently decides when to retransmit.
- Simple to implement

Cons:

1. Collisions waste slots
2. Idle slots.

2. Carrier Sense Multiple Accesses (CSMA)

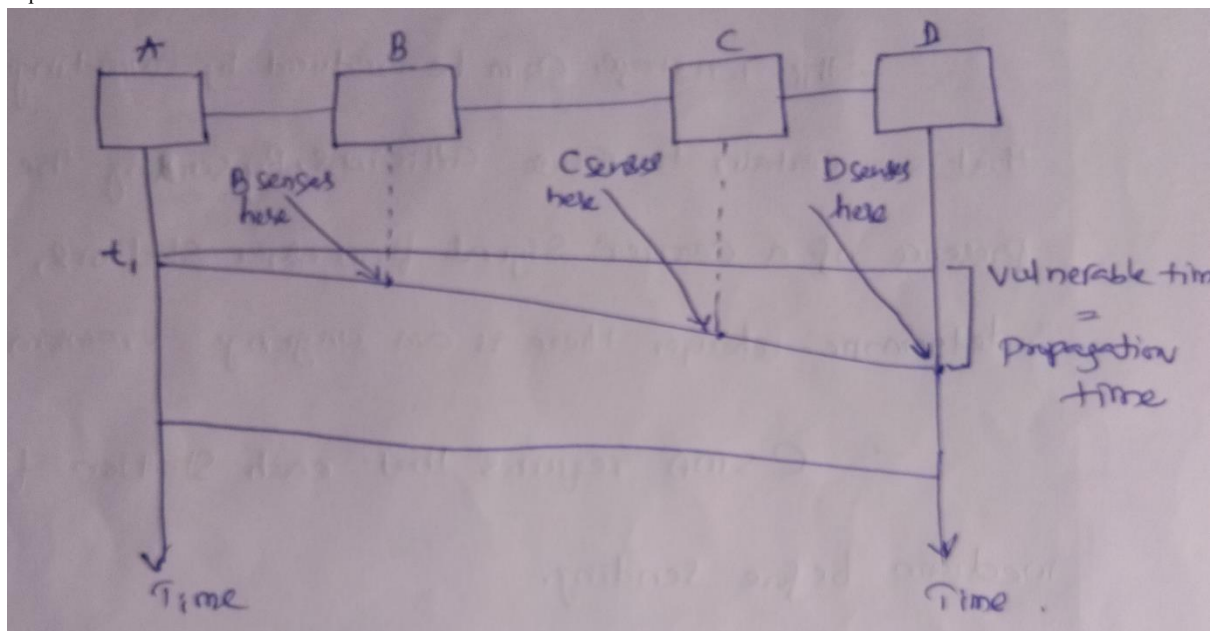
In order to minimize the chance of collision and therefore increase the performance, CSMA was developed. The low maximum through of the ALOHA schemes is due to the wastage of transmission band width because of frame collisions. This wastage can be reduced by avoiding transmissions that are certain to cause collisions. By sending the medium for the presence of a carrier signal from other stations, a station can determine whether there is an ongoing transmission.

CSMA requires that each station first listen to the medium before sending. In other words, CSMA is based on the principle “sense before transmit” or “listen before talk” CSMA can reduce the possibility of collision but it cannot eliminate it. The possibility of collision still exists because of propagation delay. A station may sense the medium and find it idle, only because the first bit sent by another station has not yet been received.

Vulnerable Time: The vulnerable time for CSMA is the propagation time (TP) this is the time needed for a signal to propagate from one end of the medium to the other.

When a station A sends a frame at time t_1 which reaches the right most station D at time $t_1 +$

T_p



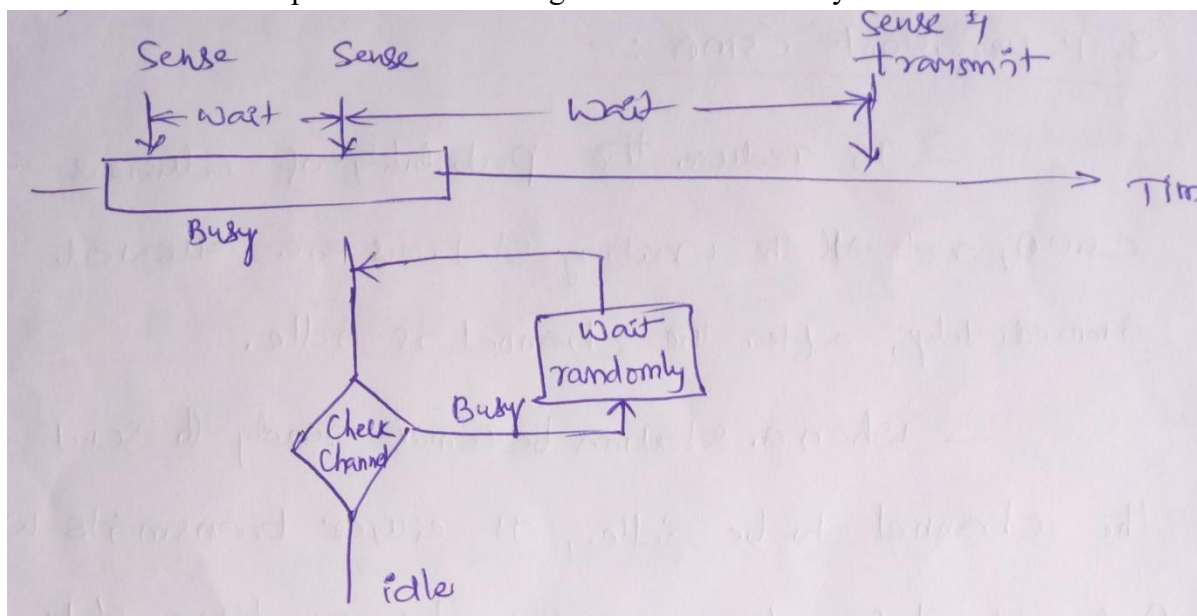
Persistence Methods: There are the three protocols

1. Non persistent CSMA
2. 1- persistent CSMA
3. P – persistent CSMA

1. Non – Persistent Method:

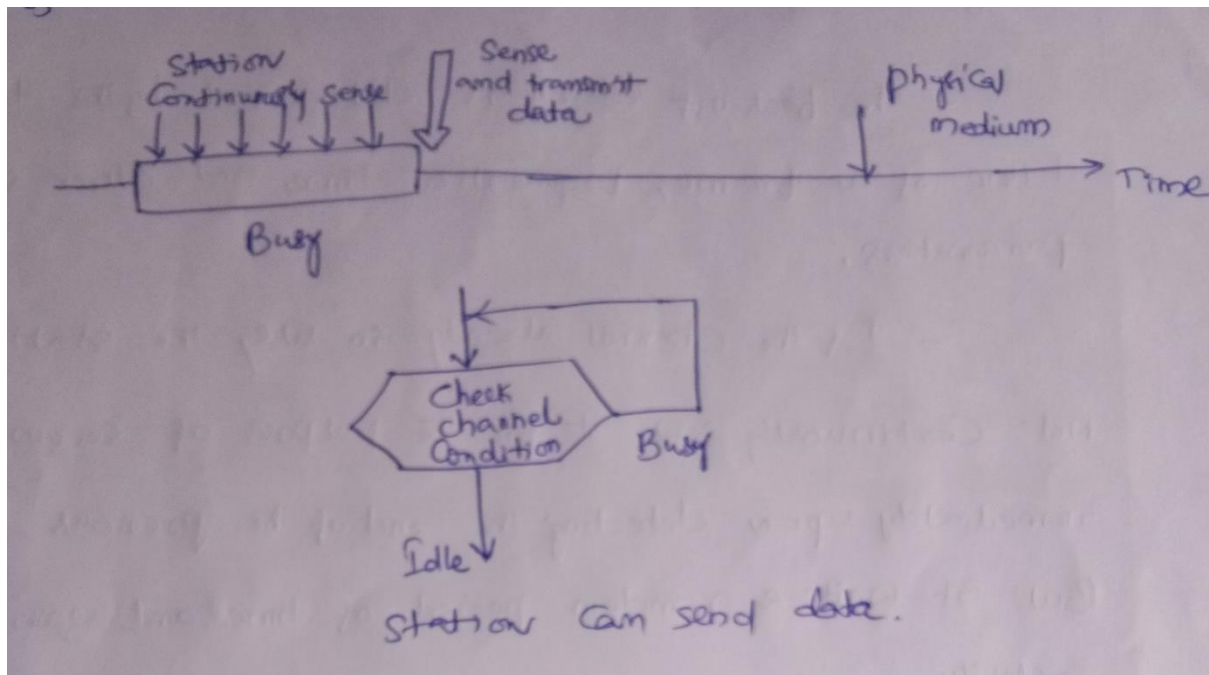
In non-persistent CSMA, when a station having a packet (frame) to transmit and finds that the channel is busy, it backs off a fixed interval of time. It then checks the channel again and if the channel is free then it transmits. The back – off delay is determined by the transmission time of a frame, propagation time and other system parameters. If the channel already in use, the stations does not continuously sense it for the purpose of seizing it immediately upon detecting the end of the previous transmission

But it waits a random period of time and again checks for activity.



2. 1- Persistent:

Any station wishing to transmit, monitor the channel continuously until the channel is idle and then transmits immediately with probability one, hence the name called 1 – persistent. When two or more stations are waiting to transmit a collision is guaranteed. Since each station will transmit immediately at the end of busy period. In this case each will wait a random amount of time and will then reattempt to transmit.

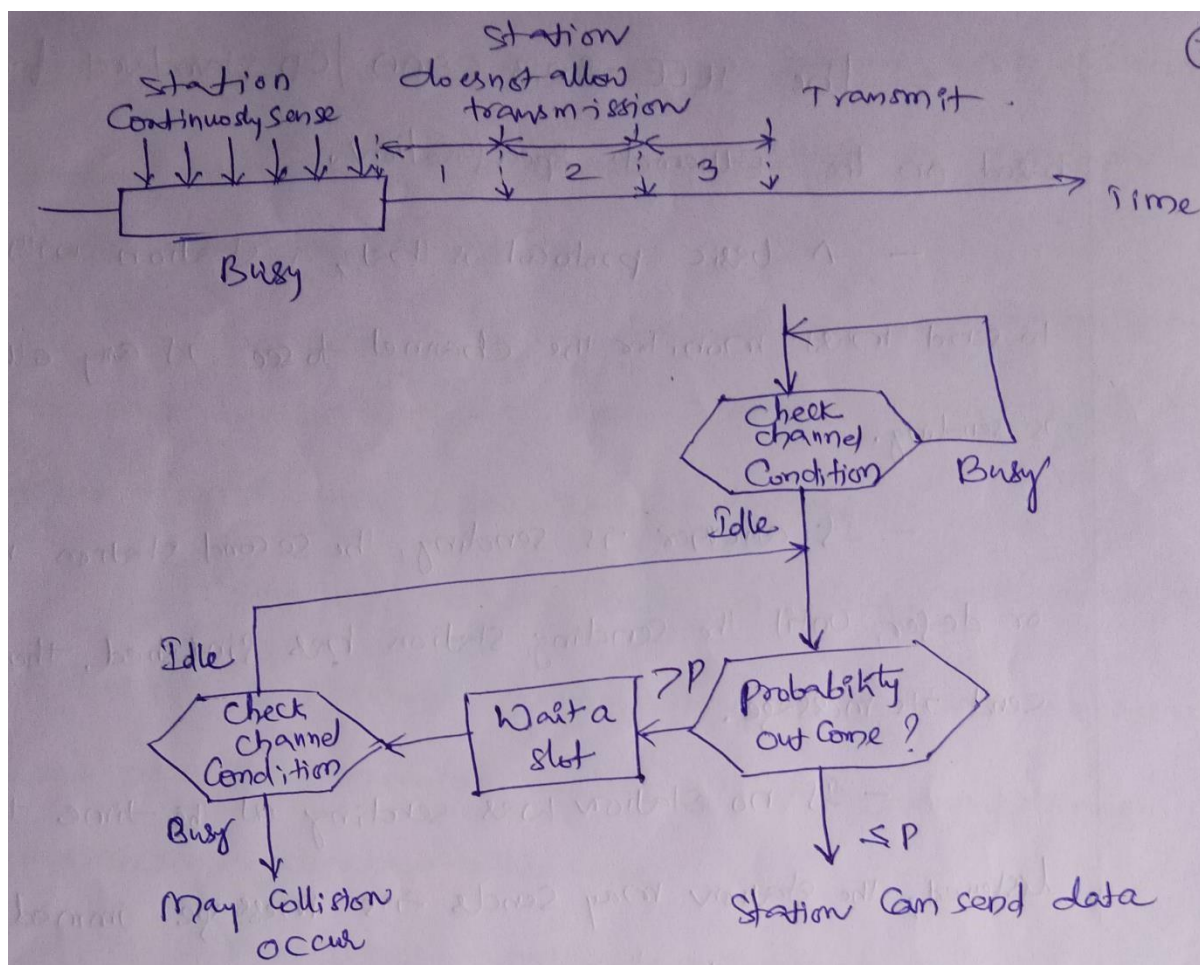


3. P – Persistent CSMA:

To reduce the probability of collisions in 1- persistent CSMA, not all the waiting stations are allowed to transmit immediately, after the channel is idle. When a station becomes ready to send and it sense the channel to be idle, it either transmits with a probability p or it defers transmission by one time slot with a probability

$$q = 1 - p$$

If deferred slot is idle, the station either transmits with probability p or defers again with a probability q . This process is repeated until either packets are transmitted or the channel becomes busy.



3.CSMA/CD (Collision Detection):

In both CSMA and ALOHA schemes, collisions involve entire frame transmission. If a station can determine whether a collision is taking place, then the amount of wasted bandwidth can be reduced by aborting the transmission when a collision is detected.

CSMA/CD is the most commonly used protocol for LANs.

It was developed jointly by digital equipment corporation (DEC), Intel and Xerox

This network is called as Ethernet. The IEEE 802.3 CSMA /CD standard for LAN based on the Ethernet specification. A basic protocol is that, a station with a message to send must monitor the channel to see if any other station is sending. If another is sending the second station must wait or defer, until the sending station has finished. Then it may send its messages. If no station was sending at the time that it first listened, the station may send its message immediately. The term "carrier sense" indicates "listening before transmitting" behaviour. If two or more stations have messages to send at the same time and they are separated by significant distance on the bus / channel, each may begin transmitting at roughly the same time without being aware of the other station. The signals from each station will superimpose on the channel and is garbled beyond the decoding ability of the receiving station. This is called "collision"

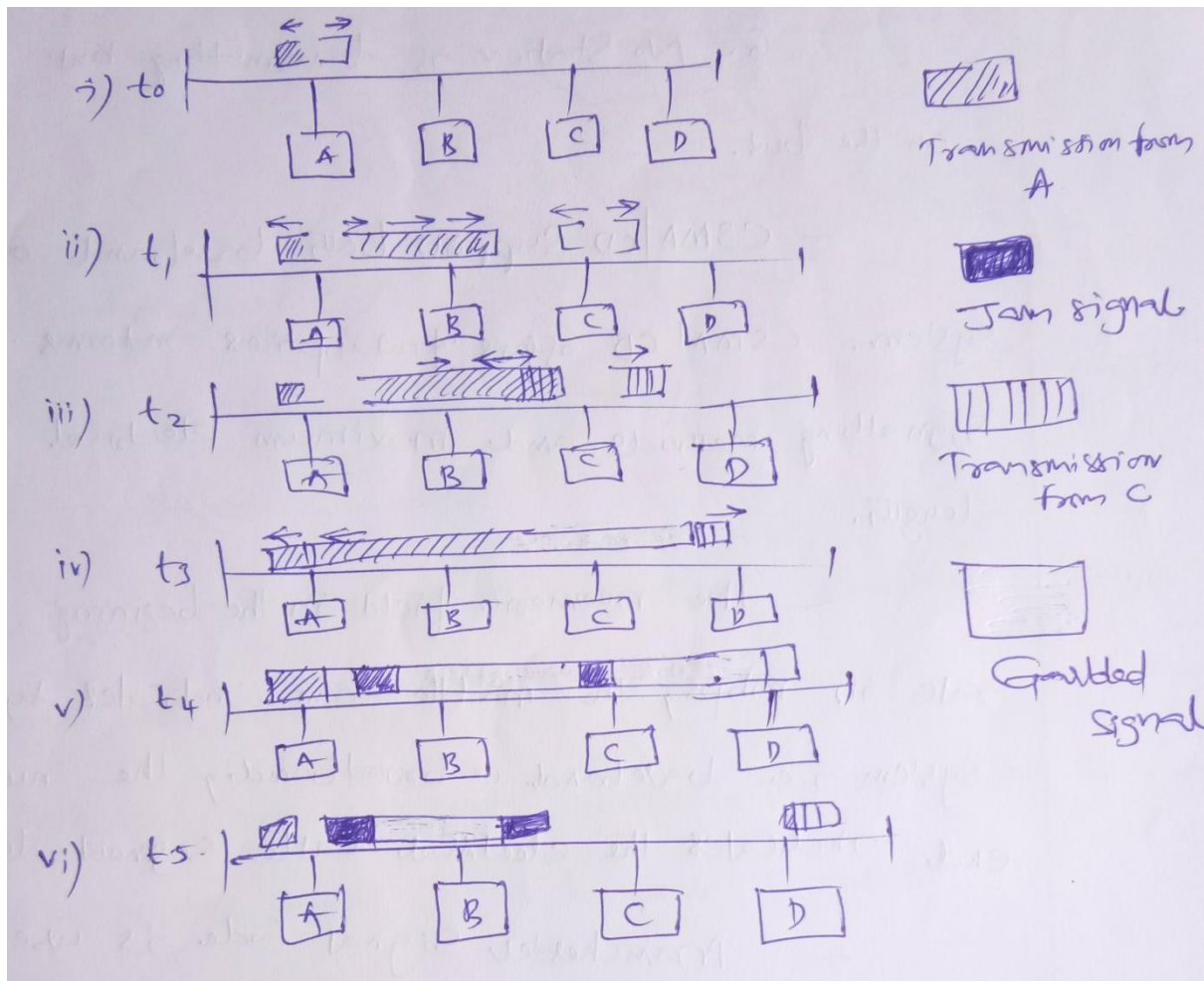
A protocol is required for transmitting station to monitor the channel while sending each of bits message and to detect such “collisions” When a collision has been detected, each of sending stations must cease transmitting wait for a random length of time, and then try again. Because of quick termination of transmission time and bandwidth is saved. Therefore CSMA/CD is more efficient then ALOHA, slotted ALOHA and CSMA.

CSMA/ CD network best on a bus, multipoint topology with bursty asynchronous transmission. All stations are attached to one path and monitor the signal on the channel through transreceiver attached to the cable.

CSMA/CD has totally decentralized control and is based on contention access.

From an above figure, station A and station D are the extreme ends of a bus structure

1. Station A listens channel starts transmitting a packet addressing D.
2. Station B and C all ready for transmission. B senses a transmission on channel so defers. C is unaware of transmission and begins its own transmission.
3. Station A transmission reaches C. C detects collision and crass transmission sends jam signal.
4. Effect of collision propagates back to A, A stops its transmission.
5. A sends jam signal.
6. No station is transmitting but there still signals on the bus.

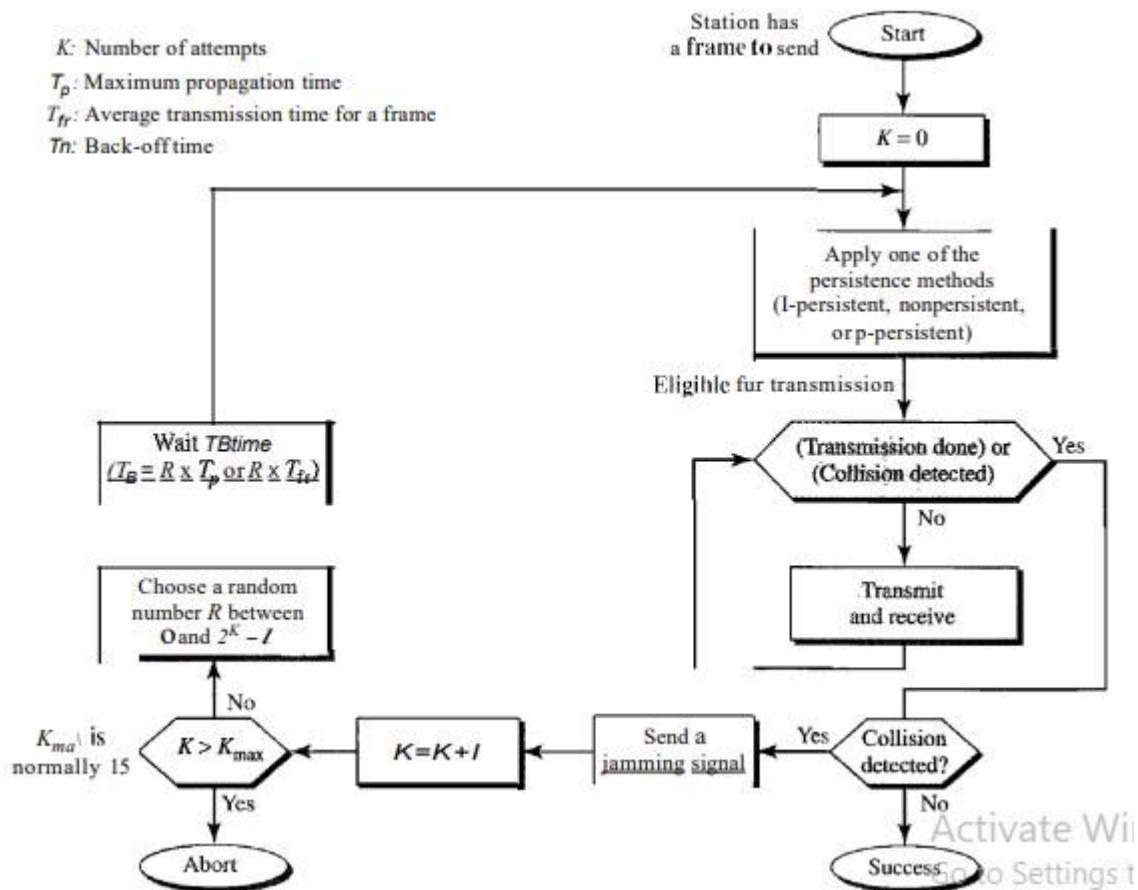


CSMA/CD supports both base band and broad band system CSMA/CD offers four options in terms of bit rate signaling methods and maximum electrical cable segment length. The numeric field in the beginning indicates the bit rate in mbps, the middle terms indicates type of signaling system i.e. back band or broad band, the numeric field in the end indicates the electrical cable segment length in $\times 100$ meters

CSMA/CD Throughput:

1. The throughput of CSMA/CD is greater than that of pure or slotted ALOHA
2. For 1- persistent method, the maximum throughput is around 50% when $G = 1$
3. For non-persistent method, the maximum throughput can go up to 90% when G is between 3 and 8.

CSMA/CD Flow Graph:



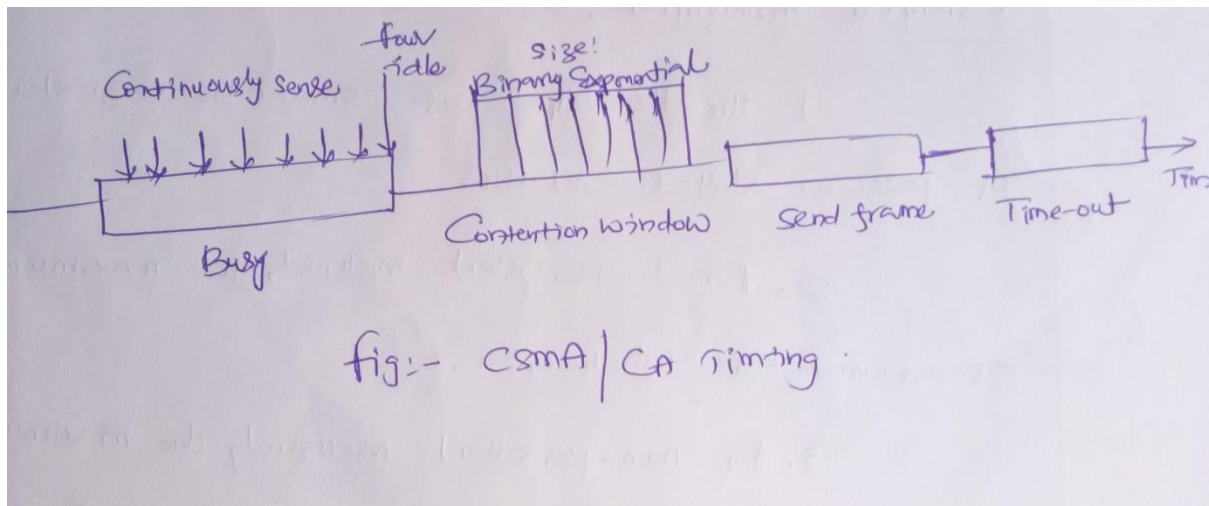
4.CSMA/CA: (collision Avoidance): In wireless n/w's cannot use CSMA/CD in the MAC sub layer, this requires the ability to receive and transmit at the same time.

In a wireless n/w's, much of the sent energy is lost transmission. The received signal has very little energy. Therefore, a collision may add only 5 to 10 percent additional energy.

This is not useful for effective collision detection; we need to avoid collision on wireless networks because they cannot be detected. Since, CSMA/CA was invented for the networks.

Collision avoidance using in three models

1. Inter frame space
2. Contention window
3. Acknowledgement



Inter Frames Space:

1. Collisions are avoided by deterring transmission even if the channel is found idle.
2. When an idle channel is found, the station does not send immediately. It waits for a period of time called the inter frame space.(IFS)
3. In CSMA/CA, the IFS can also be used to define the priority of a station of a frame. A station that is assigned shorter it's has a higher priority.

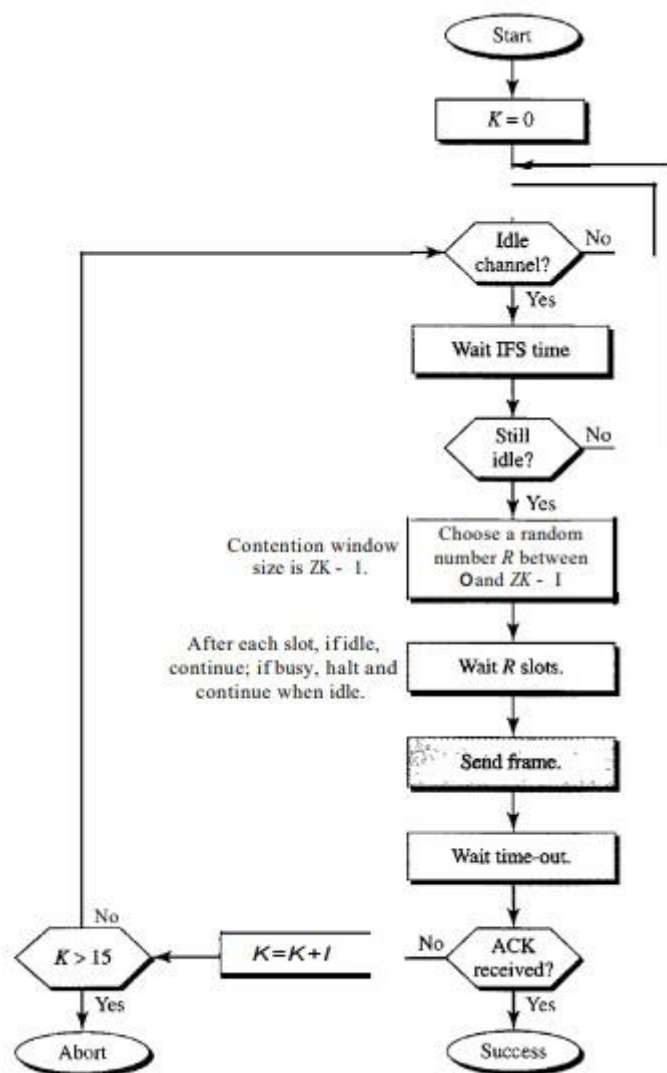
Contention Window:

1. Contention windows are an amount of time divided into slots. A station that is ready to send chooses a random no. of slots as its wait time.
2. Station set one slot for the first time and then double each time station cannot detect an idle channel after the IFS time.
3. In this method, the station needs to sense the channel after each time slot
4. If the station finds channel busy, it does not restart the process, it just stops the timer and restarts it when the channel is sensed as idle.
5. This method gives the priority to the station with the longest waiting time.

Acknowledgement:

The data may be corrupted during the transmission. The positive acknowledgement and the time out can help guarantee that the receiver has received the frame.

Flow Chart for CSMA/CA:



Difference between CSMA/CA and CSMA/CD

S.NO	CSMA/CD	CSMA/CA
1.	CSMA / CD is effective after a collision.	Whereas CSMA / CA is effective before a collision.
2.	CSMA / CD is used in wired networks.	Whereas CSMA / CA is commonly used in wireless networks.
3.	It only reduces the recovery time.	Whereas CSMA/ CA minimizes the possibility of collision.
4.	CSMA / CD resends the data frame whenever a conflict occurs.	Whereas CSMA / CA will first transmit the intent to send for data transmission.
5.	CSMA / CD is used in 802.3 standard.	While CSMA / CA is used in 802.11 standard.
6.	It is more efficient than simple CSMA(Carrier Sense Multiple Access).	While it is similar to simple CSMA(Carrier Sense Multiple Access).

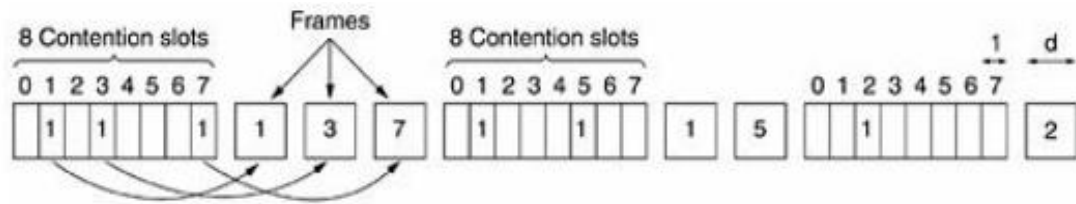
Collision Free Protocols:

Almost collisions can be avoided in **CSMA/CD**, they can still occur during the contention period. The collision during contention period adversely affects the system performance, this happens when the cable is long and length of packet are short. This problem becomes serious as fiber optics network come into use. Here we shall discuss some protocols that resolve the collision during the contention period.

- Bit-map Protocol
- Binary Countdown

Bit-map Protocol:

- In bit map protocol, the contention period is divided into N slots, where N is the total number of stations sharing the channel.
- If a station has a frame to send, it sets the corresponding bit in the slot. So, before transmission, each station knows whether the other stations want to transmit.
- If station 0 has a frame to send, it transmits a 1 bit during the zeroth slot. No other station is allowed to transmit during this slot. Regardless of what station 0 does, station 1 gets the opportunity to transmit a 1 during slot 1, but only if it has a frame queued.
- In general, station j may announce that it has a frame to send by inserting a 1 bit into slot j. After all N slots have passed by, each station has complete knowledge of which stations wish to transmit. At that point, they begin transmitting in numerical order.



- Since everyone agrees on who goes next, there will never be any collisions. After the last ready station has transmitted its frame, an event all stations can easily monitor, another N bit contention period is begun.
- If a station becomes ready just after its bit slot has passed by, it is out of luck and must remain silent until every station has had a chance and the bit map has come around again.
- Protocols like this in which the desire to transmit is broadcast before the actual transmission are called reservation protocols.

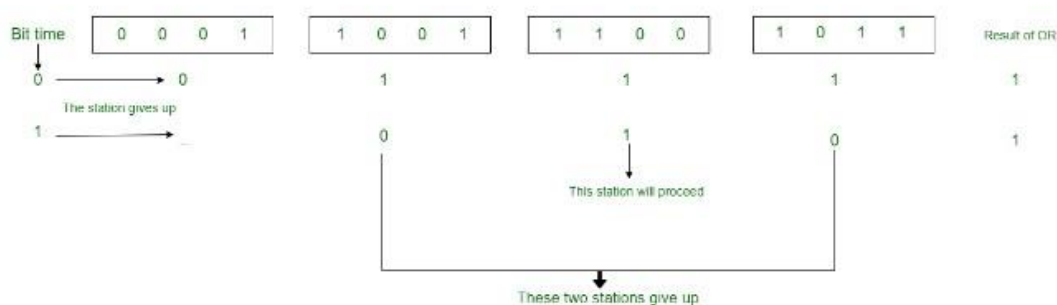
Performance of Bit – map Protocol

- High numbered stations rarely have to wait for the next scan. Since low-numbered stations must wait on average $1.5N$ slots and high-numbered stations must wait on average $0.5N$ slots, the mean for all stations is N slots.
- The channel efficiency at low load is easy to compute. The overhead per frame is N bits and the amount of data is d bits, for an efficiency of $d/(d + N)$.

Binary Countdown:

Binary countdown protocol is used to overcome the overhead 1 bit per binary station. In binary countdown, binary station addresses are used. A station wanting to use the channel broadcast its address as binary bit string starting with the high order bit. All addresses are assumed of the same length. Here, we will see the example to illustrate the working of the binary countdown. In this method, different station addresses are ORed together who decide the priority of transmitting. If these stations 0001, 1001, 1100, 1011 all are trying to seize the channel for transmission. All the station at first broadcast their most significant address bit that is 0, 1, 1, 1 respectively. The most significant bits are ORed together. Station 0001 see the 1MSB in another station addresses and knows that a higher numbered station is competing for the channel, so it gives up for the current round.

Other three stations 1001, 1100, 1011 continue. The next bit is 1 at station 1100, station 1011 and 1001 give up. Then station 110 starts transmitting a frame, after which another bidding cycle starts.

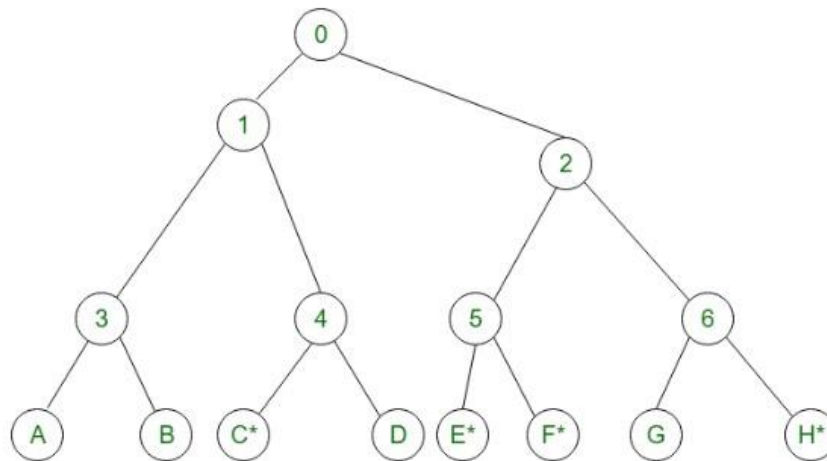


Limited Contention Protocols:

- Collision based protocols (pure and slotted ALOHA, CSMA/CD) are good when the network load is low.
- Collision free protocols (bitmap, binary Countdown) are good when load is high.
- How about combining their advantages
 1. Behave like the ALOHA scheme under light load
 2. Behave like the bitmap scheme under heavy load.

Adaptive Tree Walk Protocol:

- partition the group of station and limit the contention for each slot.
- Under light load, everyone can try for each slot like aloha □ Under heavy load, only a group can try for each slot
- **How do we do it:**
 - treat every stations as the leaf of a binary tree ○ first slot (after successful transmission), all stations can try to get the slot(under the root node).
 - if no conflict, fine
 - in case of conflict, only nodes under a subtree get to try for the next one. (depth first search)



- **Slot-0:** C*, E*, F*, H* (all nodes under node 0 can try which are going to send), conflict
- **Slot-1:** C* (all nodes under node 1 can try}, C sends
- **Slot-2:** E*, F*, H*(all nodes under node 2 can try}, conflict
- **Slot-3:** E*, F* (all nodes under node 5 can try to send), conflict
- **Slot-4:** E* (all nodes under E can try), E sends
- **Slot-5:** F* (all nodes under F can try), F sends
- **Slot-6:** H* (all nodes under node 6 can try to send), H sends.

Controlled Access Protocols

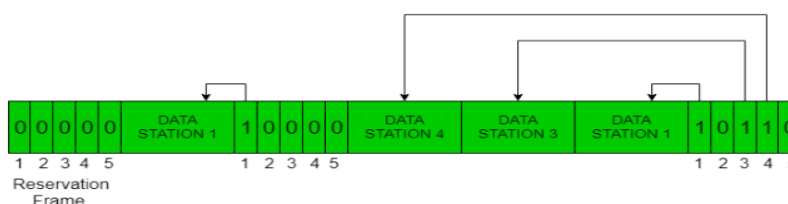
Controlled access protocols allow only one node to send data at a given time. Before initiating transmission, a node seeks information from other nodes to determine which station has the right to send. This avoids collision of messages on the shared channel.

- The station can be assigned the right to send by the following three methods–
 1. Reservation
 2. Polling
 3. Token Passing

Reservation:

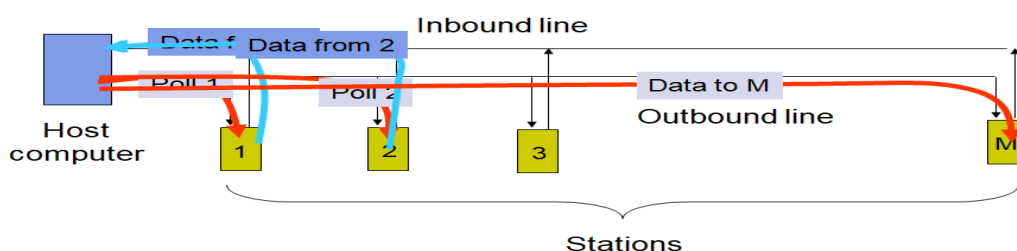
In the reservation method, a station needs to make a reservation before sending data. If there are M stations, the reservation interval is divided into M slots, and each station has one slot. Since everyone agrees on who goes next, there will never be any collisions.

- Suppose if station 1 has a frame to send, it transmits 1 bit during the slot 1. No other station is allowed to transmit during this slot.
- The stations which have reserved their slots transfer their frames in that order.
- After data transmission period, next reservation interval begins.
- The following figure shows a situation with five stations and a five slot reservation frame. In the first interval, only stations 1, 3, and 4 have made reservations. In the second interval, only station 1 has made a reservation.



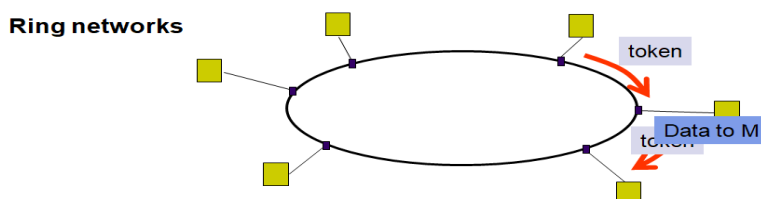
Polling:

- In this, one acts as a primary station(controller) and the others are secondary stations. All data exchanges must be made through the controller.
- The message sent by the controller contains the address of the node being selected for granting access.
- Although all nodes receive the message but the addressed one responds to it and sends data, if any. If there is no data, usually a “poll reject”(NAK) message is sent back.
- Problems include high overhead of the polling messages and high dependence on the reliability of the controller.



Token Passing:

- In token passing scheme, the stations are connected logically to each other in form of ring and access of stations is governed by tokens.
- A token is a special bit pattern or a small message, which circulate from one station to the next in the some predefined order.
- In Token ring, token is passed from one station to another adjacent station in the ring.
- Token represents permission to send. If a station has a frame queued for transmission when it receives the token, it can send that frame before it passes the token to the next station.



Channelization: Channelization is a set of methods by which the available bandwidth is divided among the different nodes for simultaneous data transfer.

The three channelization methods are–

- **Frequency division multiple access (FDMA):** The available bandwidth is divided into equal bands so that each station can be allocated its own band. Guard bands are also added so that no two bands overlap to avoid crosstalk and noise.
- **Time division multiple access (TDMA):** In this, the bandwidth is shared between multiple stations. To avoid collision time is divided into slots and stations are allotted these slots to transmit data.
- **Code division multiple access (CDMA):** Code Division Multiple Access (CDMA) – One channel carries all transmissions simultaneously. There is neither division of bandwidth nor division of time, data from different stations can be transmitted simultaneously in different code languages.

Wireless LAN's (WLAN's):

Wireless LANs are increasingly popular, and homes, offices, cafes, libraries, airports, zoos, and other public places are being outfitted with them to connect computers, PDAs, and smart phones to the Internet. Wireless LANs can also be used to let two or more nearby computers communicate without using the Internet. The main wireless LAN standard is 802.11

802.11 Architecture and protocol stack

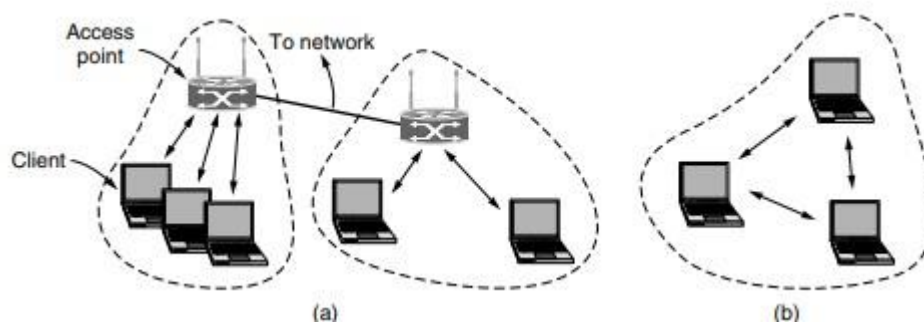


Fig. 802.11 architecture. (a) Infrastructure mode. (b) Ad-hoc mode

WLANs, as standardized by IEEE 802.11, operates in two basic modes, infrastructure, and ad hoc mode.

- **Infrastructure Mode** – Mobile devices or clients connect to an access point (AP) that in turn connects via a bridge to the LAN or Internet. The client transmits frames to other clients via the AP.
- **Ad Hoc Mode** – Clients transmit frames directly to each other in a peer-to-peer fashion.

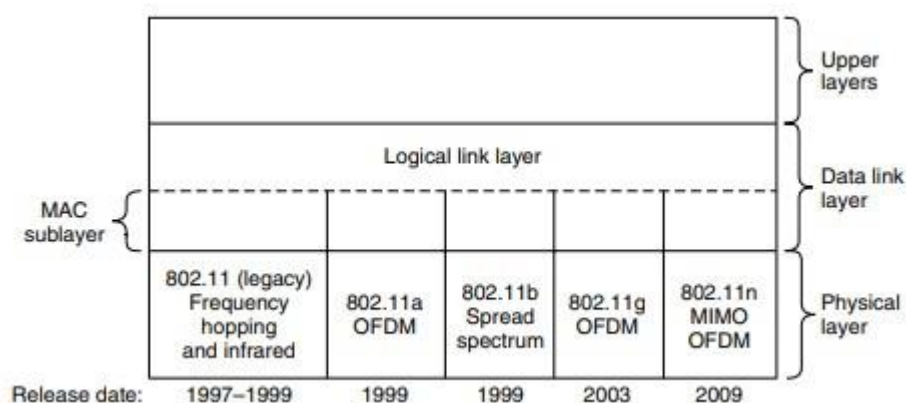


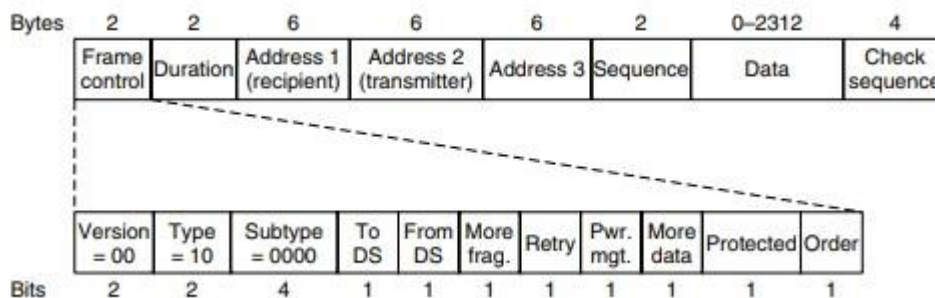
Fig. Protocol stack of 802.11

Components of WLANs

The components of WLAN architecture as laid down in IEEE 802.11 are –

- **Stations (STA)** – Stations comprises of all devices and equipment that are connected to the wireless LAN. Each station has a wireless network interface controller. A station can be of two types –
 - Wireless Access Point (WAP or AP)
 - Client
- **Basic Service Set (BSS)** – A basic service set is a group of stations communicating at the physical layer level. BSS can be of two categories –
 - Infrastructure BSS
 - Independent BSS
- **Extended Service Set (ESS)** – It is a set of all connected BSS.
- **Distribution System (DS)** – It connects access points in ESS.

802.11 Frame structure



Advantages of WLANs

- They provide clutter-free homes, offices and other networked places.
- The LANs are scalable in nature, i.e. devices may be added or removed from the network at greater ease than wired LANs.
- The system is portable within the network coverage. Access to the network is not bounded by the length of the cables.
- Installation and setup are much easier than wired counterparts. □ The equipment and setup costs are reduced.

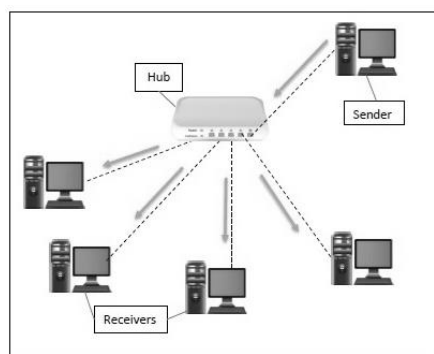
Disadvantages of WLANs

- Since radio waves are used for communications, the signals are noisier with more interference from nearby systems.
- Greater care is needed for encrypting information. Also, they are more prone to errors. So, they require greater bandwidth than the wired LANs. □ WLANs are slower than wired LANs.

Data link layer switching

Hub

A hub is a physical layer networking device which is used to connect multiple devices in a network. They are generally used to connect computers in a LAN. A hub has many ports in it. A computer which intends to be connected to the network is plugged in to one of these ports. When a data frame arrives at a port, it is broadcast to every other port, without considering whether it is destined for a particular destination or not. The process used by the Hub consumes more bandwidth and limits the amount of communication. Nowadays, the use of hub is obsolete, and it is replaced by more advanced computer network components such as Switches, Routers.



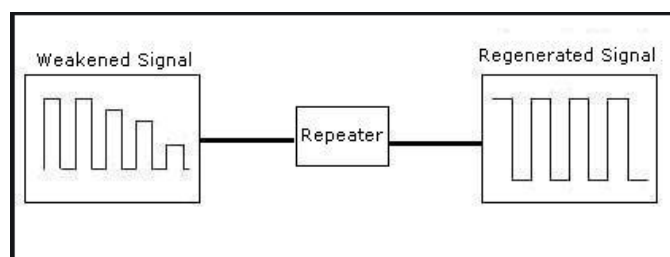
Types of Hubs used in Networking

Active hubs: Active hubs are a central connecting device in a networking that strengthens the signal, prior to sending them to the destination devices. It is also known as “multi-port repeater”. Generally, Active hubs are used to create connections in a physical star topology. It can extend maximum media distance over a network.

Passive hubs Passive hubs do not amplify the signal before broadcasting them over a network. They simply maintaining physical connection and received the frames, then rebroadcast them across all over the network. It is just a connector or connects the wires coming from different branches. The Passive hub is the collision point where the signals are come from different stations and collide.

Repeater

Repeaters are network devices, pure hardware device and operating at physical layer of the OSI model, which amplify or regenerate an incoming signal before retransmitting it. They are incorporated in networks to expand its coverage area. They are also known as signal boosters. When an electrical signal is transmitted via a channel, it gets attenuated depending upon the nature of the channel .This problem is alleviated by installing repeaters at certain intervals.



Repeaters amplify the attenuated signal and then retransmit it.

Advantages of Repeaters

- Repeaters are simple to install and can easily extend the length or the coverage area of networks.
- They are cost effective.
- Repeaters don't require any processing overhead.
- They can connect signals using different types of cables.

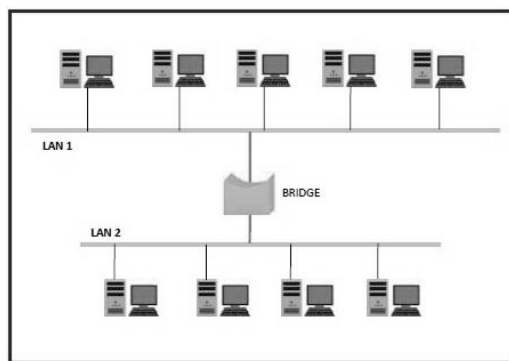
Disadvantages of Repeaters

- It is two port devices. Repeaters cannot connect dissimilar networks.
- They cannot differentiate between actual signal and noise.
- They cannot reduce network traffic or congestion.
- Most networks have limitations upon the number of repeaters that can be deployed.

Bridge

A bridge is a network device that connects multiple LANs (local area networks) together to form a larger LAN. The process of aggregating networks is called network bridging. A bridge

connects the different components so that they appear as parts of a single network. Bridges operate at the data link layer of the OSI model and hence also referred as Layer 2 switches.

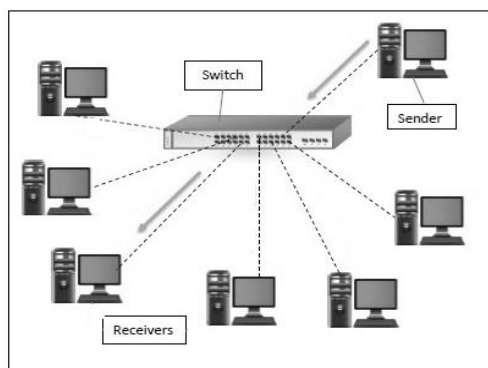


Use of Bridge in Computer Network

- A bridge in a computer network connects with other bridge networks that utilize a similar protocol. By joining multiple LANs, bridges help in multiplying the network capacity of a single LAN.
- These network devices work at the data link layer in an OSI model to connect two different networks and provide communication between them.
- Since they operate at data link layer, they transmit data as data frames. On receiving a data frame, the bridge consults a database to decide whether to pass, transmit or discard the frame.
 - If the frame has a destination MAC (media access control) address in the same network, the bridge passes the frame to that node and then discards it.
 - If the frame has a destination MAC address in a connected network, it will forward the frame toward it.
- A bridge utilizes a database to determine where to transmit or remove the data frame. In cases where the destination MAC address is not available, bridges can broadcast data frames to each node. To discover new segments, they maintain the MAC address table.
- Bridges can switch any kind of packets, this is because bridges do not examine the payload field of the data frame that arrives, but simply looks at the MAC address for switching.
- Bridges are of two types **Static Bridge** (Manual Entry to database) and **Dynamic Bridge** or **Transparent Bridge** (Automatic database update).

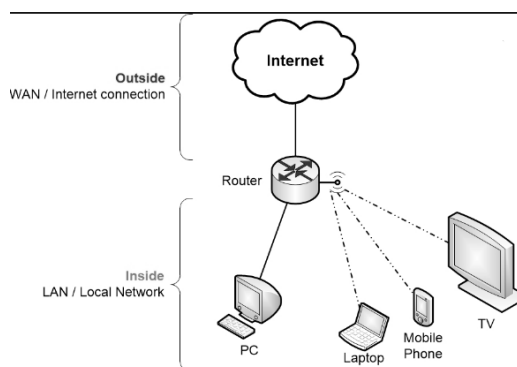
Switch

A switch is a hardware device that connects multiple devices on a computer network. A Switch contains more advanced features than Hub. The Switch contains the updated table that decides where the data is transmitted or not. Switch delivers the message to the correct destination based on the physical address present in the incoming message. A Switch does not broadcast the message to the entire network like the Hub. It determines the device to which the message is to be transmitted. Therefore, we can say that switch provides a direct connection between the source and destination. It increases the speed of the network. Thus, it supports both unicast and multicast communications.



Router

- A router is a hardware device which is used to connect a LAN with an internet connection. It is used to receive, analyze and forward the incoming packets to another network.
- A router works in a **Layer 3 (Network layer)** of the OSI Reference model.
- A router forwards the packet based on the information available in the routing table.
- It determines the best path from the available paths for the transmission of the packet.



Advantages of Router:

- **Security:** The information which is transmitted to the network will traverse the entire cable, but the only specified device which has been addressed can read the data.
- **Reliability:** If the server has stopped functioning, the network goes down, but no other networks are affected that are served by the router.
- **Performance:** Router enhances the overall performance of the network.

Modem

- A modem is a hardware device that allows the computer to connect to the internet over the existing telephone line.
- Modem is short for "Modulator-Demodulator." It converts or "modulates" an analog signal from a telephone or cable wire to digital data (1s and 0s) that a computer can recognize. Similarly, it converts digital data from a computer or other device into an analog signal that can be sent over standard telephone lines.

Gate Ways:

Gate way connects two independent networks a gate way is a protocol converter.

A gateway can accept a packet formatted for one protocol (Ex: TCP/IP) and converts it to a packet formatted for another protocol before forwarding it.

The gate way must adjust the data rate, size and data format. Gateway is generally software installed within a router.

